Manuscripts considered — mcrc-kras-a59t-pik3ca-peritoneal-pre-hipec-q8k4

Master inventory: every paper reviewed in this case

Generated 2026-05-29

Manuscripts considered — mcrc-kras-a59t-pik3ca-peritoneal-pre-

Master inventory: 61 clinical (24 included, 37 considered & excluded) + 39 pre-clinical (32 included, 7 considered & excluded) + 59 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Cha/Cha (2026) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Mass-based response drug screening to personalise HIPEC regimen
- **Indication / model:** Colorectal/appendiceal peritoneal metastases (CRS-HIPEC) (n/a); —
- **Design:** n/a
- **Effect size:** — — Feasibility, response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

FFCD/FFCD (2026) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sotorasib + panitumumab + 5-FU
- **Indication / model:** KRAS G12C-mutated mCRC, 1L (1L); KRAS G12C
- **Design:** 2
- **Effect size:** — — PFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

MGH/MGH (2026) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SBRT + botensilimab + balstilimab
- **Indication / model:** Non-MSI-H or pMMR CRC with liver metastases (2L+); MSS / pMMR; liver mets
- **Design:** early phase 1
- **Effect size:** — — Safety, ORR, abscopal effect
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

MD Anderson/MD Anderson (2026) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASCEND-CRC platform: SOC + bev / cetuximab / panitumumab / FOLFIRI rotation
- **Indication / model:** Metastatic colorectal cancer with dynamic resistance (2L+); —
- **Design:** early phase 1
- **Effect size:** — — ctDNA resistance dynamics
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Strickler/Bekaii-Saab (2025) — Nat Commun

Reference:doi:10.1038/s41467-025-67824-z · **Type:** clinical · **Inclusion:** excluded — MOUNTAINEER final analysis. Same RAS-WT exclusion as primary publication.

- **Intervention:** Tucatinib + trastuzumab (MOUNTAINEER final analysis)
- **Notes:** Excluded: MOUNTAINEER final analysis. Same RAS-WT exclusion as primary publication. — Final OS analysis — durable benefit in RAS-WT HER2+ refractory mCRC. KRAS A59T patient remains excluded by label.

Frödin/Martling (2025) — N Engl J Med

Reference:PMID 40979555 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Adjuvant low-dose aspirin in PI3K-pathway-altered resected CRC (ALASCCA)
- **Indication / model:** Resected stage I-III rectal or stage II-III colon cancer with somatic PI3K-pathway alteration (PIK3CA / PIK3R1 / PTEN) (adj); n=626 randomized; 1,103 of 2,980 screened (37%) carried a PI3K-pathway alteration. Group A = PIK3CA exon 9/20 mut; Group B = other PI3K-pathway alterations.
- **Design:** Randomized double-blind placebo-controlled phase 3
- **n:** 626
- **Effect size:** 0.49 HR (Group A, time to recurrence) — Time to recurrence at 3 years
- **Variance:** 95% CI 0.24–0.98; 95% CI 0.24-0.98 (Group A); HR 0.42 (95% CI 0.21-0.83) Group B; p=0.044 (Group A)
- **Toxicities:** GI bleeding G>=3 (3%); Serious AEs overall (5%)
- **Case fit:** partial
- **Notes:** Practice-changing 2025 phase 3 — first prospective biomarker-driven aspirin RCT in CRC to meet its primary endpoint. The patient's PIK3CA M1043I (kinase-domain, exon 20) would fall in Group A. Decision-relevant for post-HIPEC adjuvant-equivalent maintenance discussion; metastatic context limits direct extrapolation, but mechanism rationale and class safety are well-mapped.

Dumbrava/Dumbrava (2025) — Cancer Discov

Reference:doi:10.1158/2159-8290.CD-24-1304 · **Type:** clinical · **Inclusion:** excluded — Rezatapopt (PC14586, PYNNACLE) is a Y220C-pocket-specific p53 reactivator. The patient's TP53 R273 has a different structural pathology (DNA-binding domain hotspot, not the Y220C pocket); rezatapopt does not target R273. PMV's program is allele-specific by design.

- **Intervention:** Rezatapopt (PC14586, PYNNACLE) in TP53 Y220C solid tumors
- **Notes:** Excluded: Rezatapopt (PC14586, PYNNACLE) is a Y220C-pocket-specific p53 reactivator. The patient's TP53 R273 has a different structural pathology (DNA-binding domain hotspot, not the Y220C pocket); rezatapopt does not target R273. PMV's program is allele-specific by design. — Y220C-only program; R273

patient not biomarker-eligible. Listed so the board sees Y220C-targeted reactivator pharmacology exists but is allele-specific.

Khan/Bhatia (2025) — J Hematol Oncol

Reference:PMID 41213988 · **Type:** clinical · **Inclusion:** included

- **Intervention:** DT2216 BCL-xL PROTAC phase 1 first-in-human in r/r solid tumors
- **Indication / model:** Relapsed/refractory advanced solid tumors (2L+); n=20 heavily pre-treated solid tumors (mixed histologies). Median age 60.5; 60% female.
- **Design:** Phase 1 single-arm dose-escalation
- **n:** 20
- **Effect size:** 1 DLT (G4 thrombocytopenia, transient) — — Safety, RP2D, PK, target engagement (BCL-xL degradation in peripheral leukocytes)
- **Variance:** transient thrombocytopenia resolving in 3-4 days; RP2D 0.4 mg/kg IV BIW
- **Toxicities:** Thrombocytopenia G4 (1/20, 5%); Thrombocytopenia (40%); Fatigue (30%)
- **Case fit:** weak
- **Notes:** First-in-human DT2216 readout — establishes platelet-sparing profile and target engagement. No CRC-specific cohort; SD only as monotherapy. Combination cohorts (DT2216 + paclitaxel ovarian; DT2216 + irinotecan pediatric) are the relevant forward steps for the patient's BCL2L1/TOP1 amp biology. PMC12613848.

Sapience/Sapience (2025) — ASCO GI 2025 abstract / Sapience release

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** ST316 (beta-catenin/BCL9 disruptor) phase 1 in Wnt/beta-catenin-driven solid tumors
- **Indication / model:** Advanced solid tumors with Wnt/beta-catenin pathway alterations (APC, CTNNB1, AXIN1) (2L+); Phase 1 dose-escalation enrolled solid tumors likely to harbor Wnt-pathway abnormalities — CRC subset prominent.
- **Design:** Phase 1 single-arm dose-escalation
- **n:** 30
- **Effect size:** prolonged SD — — Safety, PK, target engagement (nuclear-to-cytoplasm beta-catenin shift), preliminary activity
- **Variance:** SD as primary efficacy signal; no ORR yet
- **Toxicities:** Safe and well tolerated across doses; no DLTs reported at presented doses. AE detail abstract-only.
- **Case fit:** strong
- **Notes:** ASCO GI 2025 abstract — ST316 is the most directly Wnt-axis-targeted asset in active CRC trials. APC E1295 (patient) sits in the mutation cluster region; mechanism rationale is direct. Phase 2 CRC expansion (NCT05848739) includes FOLFIRI-bev and TAS-102-bev arms aligned with the patient's likely sequence. Abstract-only; full toxicity table not yet published.

Bhatt/Jolly (2025) — bioRxiv (preprint; subsequently in Epigenetics & Chromatin 2025)

Reference:doi:10.1101/2025.07.03.662954 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TP53 R273H gain-of-function: EMT and YAP/TAZ axis in colon cancer
- **Indication / model:** HT29 and SW480 colon cancer cell lines (R273H endogenous); ChIP-seq; methylation array; gene-regulatory network simulation; in vitro invasion assays

- **Design:** Mutant p53 R273H is recruited to EMT and YAP/TAZ target promoters, altering DNA methylation and driving partial or full EMT states - a gain-of-function transcriptional program rather than simple loss of WT activity.
- **n:** n=3 biological replicates per cell line
- **Effect size:** moderate — R273H enrichment at YAP/TAZ targets correlated with hypomethylation, partial-EMT signatures, and increased migratory potential in HT29 cells. Connects the patient's R273 mutation to peritoneal-spread biology (EMT) and helps justify why TP53-R273-mutant CRC may be uniquely aggressive in peritoneal disease.
- **Variance:** vs isogenic p53-null and WT-p53 controls; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line and computational only - no in vivo regression study. The R273H specificity and CRC context make it directly relevant for the patient's biology even at low translatability.

Tsai/Sutton (2025) — ACS Med Chem Lett (discovery) / Cancer Discov 2025 (mechanism)

Reference:PMID 39773897 · **Type:** preclinical · **Inclusion:** excluded — Rezatapopt is Y220C-selective; the patient's TP53 hotspot is R273, not Y220C. Reactivation pocket is not present in R273 mutants. Listed for audit trail because rezatapopt is on the trial list and the exclusion logic should be visible.

- **Intervention:** Rezatapopt (PC14586) Y220C-selective p53 reactivator
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Rezatapopt is Y220C-selective; the patient's TP53 hotspot is R273, not Y220C. Reactivation pocket is not present in R273 mutants. Listed for audit trail because rezatapopt is on the trial list and the exclusion logic should be visible.

Martling/Bjork (2025) — N Engl J Med

Reference: — · **Type:** preclinical · **Inclusion:** excluded — ALASCCA itself is a clinical RCT, not preclinical; cited via the clinical_evidence track. The mechanistic preclinical underpinning (Liao 2012 + Gu 2019 PIK3CA-aspirin in vitro) is already captured in pe-aspirin-pik3ca-liao-2012. Avoiding duplication.

- **Intervention:** ALASCCA RCT - clinical, not preclinical
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: ALASCCA itself is a clinical RCT, not preclinical; cited via the clinical_evidence track. The mechanistic preclinical underpinning (Liao 2012 + Gu 2019 PIK3CA-aspirin in vitro) is already captured in pe-aspirin-pik3ca-liao-2012. Avoiding duplication.

Cashin/Cashin (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** HIPEC with 5-FU, irinotecan, or oxaliplatin after cytoreductive surgery
- **Indication / model:** Colorectal cancer with peritoneal metastases (CRS + HIPEC) (n/a); —
- **Design:** 3
- **Effect size:** — — OS, DFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Raoof/Raoof (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pressurized intraperitoneal aerosol chemotherapy (PIPAC) with oxaliplatin or 5-FU + concurrent systemic chemo
- **Indication / model:** Peritoneal carcinomatosis from CRC and appendiceal cancer (PIPAC) (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, peritoneal regression grade
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Foo/Foo (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** PIPAC
- **Indication / model:** Peritoneal surface malignancies (CRC included) (2L+); —
- **Design:** 2
- **Effect size:** — — Safety, response, QoL
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Cloyd/Cloyd (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Flat-dose mitomycin C HIPEC vs weight-based mitomycin C HIPEC
- **Indication / model:** Colorectal peritoneal carcinomatosis (CRS-HIPEC) (n/a); —
- **Design:** 2
- **Effect size:** — — PK (AUC, clearance, half-life)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Briceño/Briceño (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Neoadjuvant chemo ± HIPEC then surgery
- **Indication / model:** Locally advanced CRC with peritoneal metastases (neoadjuvant) (neoadj); —
- **Design:** 3
- **Effect size:** — — OS, DFS
- **Case fit:** partial

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Revolution Medicines/Revolution Medicines (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** RMC-6236 (daraxonrasib) ± bevacizumab or cetuximab + FOLFOX/FOLFIRI; RMC-9805 for G12D arms
- **Indication / model:** RAS-mutated GI solid tumors (CRC, PDAC) — RASolve GI platform (2L+); RAS mutation (subprotocols A/B accept any RAS-mutated CRC; D/E/F gate on KRAS G12D)
- **Design:** 1/2
- **Effect size:** — — Safety, ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

JOYO Pharma/JOYO Pharma (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JYP0015 (pan-RAS inhibitor)
- **Indication / model:** Advanced solid tumors (PDAC, NSCLC, CRC) with RAS mutations (2L+); KRAS/NRAS/HRAS at codons 12, 13, 61, 117, or 146
- **Design:** 1/2
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Jacobio/Jacobio (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** JAB-23E73
- **Indication / model:** Advanced solid tumors with KRAS alterations (2L+); KRAS alteration (codon-agnostic by protocol text)
- **Design:** 1/2
- **Effect size:** — — Safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Servier/Servier (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** S241656 (ERK inhibitor) ± FOLFOX/FOLFIRI/cetuximab/panitumumab

- **Indication / model:** RAS/MAPK mutation-positive solid tumors (NSCLC, CRC, PDAC, BTC, histiocytosis) (2L+); KRAS, HRAS, NRAS, BRAF or CRAF mutations or alterations (codon-agnostic)
- **Design:** 1/2
- **Effect size:** — — Safety, ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Petrov NMRC/Petrov NMRC (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Leflunomide OR (MEK inhibitor + hydroxychloroquine ± bevacizumab)
- **Indication / model:** Refractory RAS-mutated metastatic solid tumors (CRC, PDAC, NSCLC, melanoma) (3L+); Documented RAS (KRAS, HRAS, NRAS) mutation — codon-agnostic
- **Design:** 2
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Elicio/Elicio (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ELI-002 7P amphiphile peptide vaccine
- **Indication / model:** KRAS/NRAS-mutated PDAC and CRC (adjuvant/MRD) (adj); KRAS G12D, G12R, G12V, G12A, G12C, G12S, G13D mutations
- **Design:** 1/2
- **Effect size:** — — MRD clearance, RFS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Johns Hopkins/Zheng (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** KRAS peptide vaccine + Poly-ICLC + balstilimab + botensilimab
- **Indication / model:** Stage IV MMR-proficient CRC and PDAC (2L+); One of the KRAS mutations included in the vaccine (G12C/G12D/G12V/G12R/G13D types)
- **Design:** 1
- **Effect size:** — — Safety, immune response, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hoffmann-La Roche/Hoffmann-La Roche (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Inavolisib + cetuximab; Inavolisib + bevacizumab; Divarasib + cetuximab ± FOLFOX/FOLFIRI; atezolizumab combos
- **Indication / model:** Metastatic colorectal cancer (INTRINSIC platform) (2L+); Cohort-specific (PIK3CA-mutant for inavolisib arms; KRAS G12C for divarasib arms)
- **Design:** 1
- **Effect size:** — — Safety, ORR, PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Strickler/Strickler (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tucatinib + trastuzumab + mFOLFOX6
- **Indication / model:** HER2+ RAS-WT metastatic colorectal cancer (MOUNTAINEER-03, 1L) (1L); HER2 IHC 3+ or ISH amplified OR NGS amplification AND RAS wild-type
- **Design:** 3
- **Effect size:** — — PFS, OS
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Emory/Emory (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Zanidatamab pre-op
- **Indication / model:** HER2+ colon/rectal cancer (neoadjuvant) (neoadj); HER2-positive
- **Design:** 2
- **Effect size:** — — pCR, MPR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Suncadia/Suncadia (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SHR-A1811 (HER2 ADC)
- **Indication / model:** Advanced CRC after standard therapy (3L+); —
- **Design:** 3
- **Effect size:** — — OS, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

GONO/GONO (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Trastuzumab deruxtecan + capecitabine + bevacizumab
- **Indication / model:** HER2+ metastatic CRC (CHIMERA) (1L); HER2 IHC 3+ or 2+/ISH amplified
- **Design:** 2
- **Effect size:** — — PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

PMV Pharmaceuticals/PMV Pharmaceuticals (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Rezatapopt (PC14586) ± pembrolizumab
- **Indication / model:** Advanced solid tumors with TP53 Y220C mutation (PYNNACLE) (2L+); TP53 Y220C mutation only
- **Design:** 1/2
- **Effect size:** — — ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Stover/Stover (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DT2216 (BCL-xL PROTAC) + paclitaxel
- **Indication / model:** Platinum-resistant ovarian cancer (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Children's Oncology Group/Children's Oncology Group (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DT2216 + irinotecan
- **Indication / model:** Relapsed/refractory pediatric solid tumors (2L+); —
- **Design:** 1/2
- **Effect size:** — — Safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Genentech/Genentech (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Divarasib (GDC-6036) ± atezolizumab / cetuximab / bevacizumab / erlotinib
- **Indication / model:** KRAS G12C-mutated NSCLC and CRC (2L+); KRAS G12C
- **Design:** 1
- **Effect size:** — — Safety, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Canadian Cancer Trials Group/CCTG (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab
- **Indication / model:** Chemo-refractory unresectable MSS metastatic CRC (3L+); MSS / pMMR
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

USC/USC (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab + fasting-mimicking diet + vitamin C
- **Indication / model:** KRAS-mutant MSS metastatic CRC (2L+); MSS + any KRAS mutation
- **Design:** 1
- **Effect size:** — — Safety, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

City of Hope/City of Hope (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab + regorafenib
- **Indication / model:** MSS metastatic colorectal cancer (2L+); MSS
- **Design:** 1/2
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Agenus/Agenus (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab
- **Indication / model:** Metastatic colorectal cancer (MSS) (2L+); MSS
- **Design:** 2
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

City of Hope/City of Hope (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FOLFOX + bevacizumab + botensilimab + balstilimab (3B-FOLFOX)
- **Indication / model:** Metastatic MSS colorectal adenocarcinoma (1L); MSS
- **Design:** 1/2
- **Effect size:** — — ORR, PFS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Agenus/Agenus (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab (expanded access)
- **Indication / model:** CRC and PDAC pre-approval access (2L+); —
- **Design:** n/a
- **Effect size:** — — Access
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Stingray/Stingray (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SR-8541A (ENPP1 inhibitor) + botensilimab + balstilimab
- **Indication / model:** Refractory MSS metastatic CRC (3L+); MSS
- **Design:** 2
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

Huazhong Univ/Huazhong Univ (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Cadonilimab (PD-1/CTLA-4 bispecific) + fruquintinib + SBRT
- **Indication / model:** Metastatic colorectal cancer (3L+) (3L+); —
- **Design:** 2
- **Effect size:** — — ORR, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Weill Cornell/Weill Cornell (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab + AGEN1423 + radiation
- **Indication / model:** CRC with liver metastases (2L+); —
- **Design:** 2
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Chipscreen/Chipscreen (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tucidinostat (HDAC inhibitor) + sintilimab + bevacizumab
- **Indication / model:** MSS / pMMR colorectal cancer (2L+); MSS / pMMR
- **Design:** 3
- **Effect size:** — — OS, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Novimmune/Novimmune (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NILK-2301 (CEA TCB bispecific)
- **Indication / model:** Locally advanced or metastatic low-tumor-volume CRC (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

IRST/IRST (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pembrolizumab + autologous dendritic cell vaccine followed by TAS-102 + bev
- **Indication / model:** MSS metastatic colorectal cancer (sequential immunochemotherapy) (2L+); MSS
- **Design:** 2
- **Effect size:** — — ORR, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

NRG Oncology/NRG Oncology (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ctDNA-guided adjuvant chemo (mFOLFOX6, CAPOX, mFOLFIRINOX)
- **Indication / model:** Resected stage II/III/IV colon cancer (CIRCULATE-US / NRG-GI008) (adj); Signatera ctDNA-positive vs negative arms
- **Design:** 2/3
- **Effect size:** — — DFS, OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

MD Anderson/MD Anderson (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ctDNA-directed neoadjuvant chemo
- **Indication / model:** Appendiceal / colorectal adenocarcinoma (ctDNA-directed neoadjuvant) (neoadj); ctDNA detectable
- **Design:** n/a
- **Effect size:** — — Feasibility, response
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Sapience/Sapience (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ST316 (Wnt/beta-catenin/STAT3 disruptor)
- **Indication / model:** Advanced metastatic CRC and other solid tumors (2L+); STAT3 / beta-catenin pathway alterations (APC, CTNNB1, AXIN1)
- **Design:** 1

- **Effect size:** — — Safety, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Huang/Huang (2025) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Glumetinib + fruquintinib
- **Indication / model:** MET amplification or overexpression mCRC (3L+) (3L+); MET amplification or protein overexpression
- **Design:** 1/2
- **Effect size:** — — ORR, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Rovers/de Hingh (2024) — Lancet Oncol / ASO

Reference:PMID 39550351 · **Type:** clinical · **Inclusion:** excluded — Perioperative systemic chemo (FOLFOX/CAPOX/FOLFIRI ± bev) around CRS-HIPEC. Backbone chemo + bev; mechanism does not target the patient's features. Relevant as the sequencing-with-HIPEC trial for the patient's window.

- **Intervention:** Perioperative systemic chemo + CRS-HIPEC vs CRS-HIPEC alone (CAIRO6)
- **Notes:** Excluded: Perioperative systemic chemo (FOLFOX/CAPOX/FOLFIRI ± bev) around CRS-HIPEC. Backbone chemo + bev; mechanism does not target the patient's features. Relevant as the sequencing-with-HIPEC trial for the patient's window. — Phase 2/3 perioperative therapy improved PFS/DFS but not OS over CRS-HIPEC alone in resectable CPM; safety acceptable. Decision-relevant for the patient's planned HIPEC pathway. Audit-trail row.

Raghav/Yoshino (2024) — Lancet Oncol

Reference:PMID 39116902 · **Type:** clinical · **Inclusion:** excluded — HER2+ ADC, including RAS-mutant patients (unlike MOUNTAINEER). HER2 is not among the patient's nominated features; pending NGS would gate eligibility. Out of Libby scope at intake.

- **Intervention:** Trastuzumab deruxtecan (DESTINY-CRC02, HER2+ mCRC, 5.4 vs 6.4 mg/kg)
- **Notes:** Excluded: HER2+ ADC, including RAS-mutant patients (unlike MOUNTAINEER). HER2 is not among the patient's nominated features; pending NGS would gate eligibility. Out of Libby scope at intake. — ORR 38% (5.4 mg/kg arm), mPFS 5.8 mo, mOS ~16 mo. RAS-mutant subset allowed, including KRAS-mutated patients — would become highly relevant if HER2 returns amplified on pending NGS. ILD ~8% (5.4 mg/kg).

Kuboki/Fakih (2024) — J Clin Oncol

Reference:doi:10.1200/JCO-24-02026 · **Type:** clinical · **Inclusion:** excluded — CodeBreaK 300 OS update — same KRAS G12C-only eligibility as primary publication. Not in mechanism scope for KRAS A59T.

- **Intervention:** Sotorasib + panitumumab OS update (CodeBreaK 300)
- **Notes:** Excluded: CodeBreaK 300 OS update — same KRAS G12C-only eligibility as primary publication. Not in mechanism scope for KRAS A59T. — OS update favored sotorasib 960 mg arm numerically. Audit-trail row.

Li/Govindan (2024) — Ann Oncol / ASCO 2024

Reference: — · **Type:** clinical · **Inclusion:** excluded — CodeBreaK 101 expansion — sotorasib + pembrolizumab / atezolizumab. KRAS G12C-only eligibility.

- **Intervention:** Sotorasib + ICI combinations (CodeBreaK 101)
- **Notes:** Excluded: CodeBreaK 101 expansion — sotorasib + pembrolizumab / atezolizumab. KRAS G12C-only eligibility. — Sotorasib + ICI mainly NSCLC. Audit-trail row only — A59T not eligible.

Garassino/Jänne (2024) — ESMO 2024 / WCLC 2024

Reference: — · **Type:** clinical · **Inclusion:** excluded — KRYSTAL-7 adagrasib + pembrolizumab in KRAS G12C NSCLC. G12C-only; A59T patient is not eligible by allele or by tumor type.

- **Intervention:** Adagrasib + pembrolizumab (KRYSTAL-7, KRAS G12C NSCLC)
- **Notes:** Excluded: KRYSTAL-7 adagrasib + pembrolizumab in KRAS G12C NSCLC. G12C-only; A59T patient is not eligible by allele or by tumor type. — KRAS G12C NSCLC; not CRC and not A59T-relevant.

Jhaveri/Turner (2024) — N Engl J Med

Reference:PMID 39476340 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Inavolisib + palbociclib + fulvestrant (INAVO120, HR+/HER2- PIK3CA-mut breast cancer)
- **Indication / model:** HR+/HER2- PIK3CA-mutated metastatic breast cancer (1L after endocrine therapy) (1L); n=325 with PIK3CA alterations confirmed by ctDNA or tissue.
- **Design:** Randomized double-blind phase 3
- **n:** 325
- **Effect size:** 0.43 HR — Investigator-assessed PFS
- **Variance:** 95% CI 0.32–0.59; 95% CI 0.32-0.59; p=<0.001
- **Toxicities:** Hyperglycemia G>=3 (5.6%); Hyperglycemia (58.6%); Stomatitis G>=3 (5.6%); Diarrhea G>=3 (3.7%); Rash (25%); Treatment discontinuation due to AE (6.5%)
- **Case fit:** cross_tumor_only
- **Notes:** Cross-tumor anchor for inavolisib's PIK3CA-mutant pharmacology — drove the Oct 2024 FDA approval in HR+/HER2- breast cancer. INTRINSIC mCRC platform (NCT04929223) extends inavolisib + bev into PIK3CA-mut mCRC; cross-tumor activity here is the strongest mechanistic basis.

Yang/Meyerhardt (2024) — J Clin Oncol

Reference:PMID 38889377 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Adjuvant celecoxib in PIK3CA-activated stage III colon cancer (CALGB/SWOG 80702)
- **Indication / model:** Resected stage III colon cancer, PIK3CA-activating subset (adj); Pre-specified biomarker subgroup of CALGB/SWOG 80702 (n=2,524 randomized total; ~16% PIK3CA-activating).
- **Design:** Pre-specified biomarker subgroup of randomized double-blind phase 3

- **n:** 401
- **Effect size:** 0.56 HR (DFS, PIK3CA-activated) — DFS and OS in PIK3CA-activating subset
- **Variance:** 95% CI 0.35–0.88; 95% CI 0.35-0.88; OS HR 0.41 (95% CI 0.21-0.79); p=0.012
- **Toxicities:** Celecoxib G3+ AEs class-typical (cardiovascular, GI). Cohort safety dominated by FOLFOX backbone toxicity.
- **Case fit:** partial
- **Notes:** Pre-specified PIK3CA-activating subset of CALGB/SWOG 80702 — 56% DFS HR and 41% OS HR favoring celecoxib are large effects, despite the overall trial being negative. Mechanism parallels the aspirin signal (COX-2 → PGE2 → PI3K/AKT/mTOR axis). Decision-relevant for combination strategy with the patient's PIK3CA M1043I.

Pulvermacher/Pulvermacher (2024) — AACR 2024 abstract (CT146)

Reference:doi:10.1158/1538-7445.AM2024-CT146 · **Type:** clinical · **Inclusion:** included

- **Intervention:** AZD0466 dual BCL-2/BCL-xL inhibitor (dendrimer-conjugated AZD4320) phase 1
- **Indication / model:** Advanced hematologic malignancies and solid tumors at varying TLS risk (2L+); Phase 1 dose-escalation across hematologic malignancies (NHL, MM) and solid-tumor low-TLS-risk cohort. Solid-tumor expansion limited.
- **Design:** Phase 1 single-arm dose-escalation
- **n:** 50
- **Effect size:** tolerable to date — — Safety, RP2D, PK
- **Variance:** no DLTs reported in abstract; dose-escalation ongoing
- **Toxicities:** Dendrimer conjugation mitigates the cardiovascular signal of free AZD4320. No DLTs in early dose-escalation per abstract; on-target thrombocytopenia monitored.
- **Case fit:** weak
- **Notes:** Abstract-only AACR 2024 readout — full safety / activity table not yet published. Mechanism rationale for BCL2L1-amp tumors is mixed because AZD0466 is dual Bcl-2/Bcl-xL rather than BCL-xL-selective. Audit-trail row for the BCL-xL class landscape. AE detail not in abstract; full toxicity table not in PMC; no published manuscript yet.

Bullock/Gordon (2024) — Nat Med

Reference:PMID 38871975 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Botensilimab + balstilimab phase 1 in r/r MSS mCRC
- **Indication / model:** Relapsed/refractory MSS metastatic colorectal cancer (2L+); n=101 r/r MSS mCRC, mixed liver / non-liver metastatic distribution. Most patients had ≥3 prior lines.
- **Design:** Phase 1b expansion cohort, single-arm
- **n:** 101
- **Effect size:** 17 % — Confirmed ORR, mOS, mPFS by liver-metastasis status
- **Variance:** 95% CI 10–26; 95% CI 10-26%; NLM subset ORR 22%, LM subset ORR 5%
- **Toxicities:** Diarrhea/colitis G>=3 (12%); Fatigue (45%); Pyrexia (38%); Hepatitis (transaminitis) G>=3 (8%); Treatment-related AE >=3 G>=3 (35%); Treatment-related death G5 (0/101, 0%)
- **Case fit:** strong
- **Notes:** Most decision-relevant MSS-CRC ICI signal in the modern era — Fc-enhanced CTLA-4 (botensilimab) + PD-1 (balstilimab) delivered a real signal in MSS mCRC, especially in no-liver-mets patients. The patient's peritoneal-only disease post-liver-met resection is the NLM phenotype. Strongest MSS-CRC immunotherapy

candidate at 2L+. PMC11405281.

Bullock/Bullock (2024) — J Clin Oncol (ASCO GI / ESMO GI 2024 abstracts)

Reference:doi:10.1200/JCO.2025.43.4_suppl.23 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Botensilimab ± balstilimab in MSS mCRC NLM (randomized phase 2 update)
- **Indication / model:** Refractory MSS metastatic colorectal cancer without active liver metastases (3L+); n=123 MSS mCRC NLM, randomized to BOT mono vs BOT+BAL. Peritoneum, soft-tissue, and brain metastases all responded.
- **Design:** Randomized open-label phase 2 (Agenus C-1101)
- **n:** 123
- **Effect size:** 19 % — ORR, mOS, peritoneal-site response
- **Variance:** BOT+BAL ORR 19% confirmed; BOT mono lower
- **Toxicities:** Diarrhea/colitis G>=3 (13%); Pyrexia/cytokine-release-like (40%); Treatment-related AE >=3 G>=3 (38%)
- **Case fit:** strong
- **Notes:** ASCO GI 2025 / ESMO GI 2024 randomized phase 2 update — peritoneal-site response specifically called out, which is the patient's dominant disease compartment. CCTG phase 3 (NCT07152821, in trials.jsonl) is the next confirmatory step. Strongest current MSS-CRC ICI-combo signal. Toxicity rates from ASCO abstract; full per-term breakdown pending peer-reviewed publication.

Jiang/Singh (2024) — Nature

Reference:PMID 38778099 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Daraxonrasib (RMC-6236) RAS(ON) multi-selective inhibitor
- **Indication / model:** KRAS-mutant PDAC + NSCLC + CRC PDX, syngeneic allograft, GEMM, organoid panels; covered G12X, G13D, Q61X variants
- **Design:** Sanglifehrin-derived molecular glue that forms a tri-complex with cyclophilin A and GTP-bound RAS (RAS(ON)) across mutant + WT isoforms, occluding effector binding.
- **n:** multiple model series; group sizes 6-10/arm typical
- **Effect size:** strong — Tumor regression at 25-50 mg/kg PO QD across PDAC, NSCLC, CRC PDX carrying G12D/V/R, G13D and Q61 alleles, and rescue of KRAS-G12C-inhibitor-resistant models via WT-RAS suppression. A59-specific in-vivo data are not in the headline package, so this read across is structural plausibility, not direct evidence.
- **Variance:** vs vehicle + KRAS-G12C-selective comparators (sotorasib/adagrasib) in resistance models; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Companion Nature paper (Knox et al., PMID 38778097) gives the PDAC GEMM regression data. The atypical-allele coverage is inferred from mechanism (binds RAS(ON) regardless of mutation site that does not disrupt the binding interface); confirm A59 sensitivity in any biochemical add-on data Revolution Medicines has released.

Knox/Stokoe (2024) — Nature

Reference:PMID 38778097 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** RMC-7977 RAS(ON) multi-selective tool compound (preclinical analog of RMC-6236)
- **Indication / model:** PDAC GEMMs (KPC, KPF), syngeneic and PDX models; broad CRC and NSCLC xenografts
- **Design:** Same RAS(ON) tri-complex mechanism as RMC-6236; selectively inhibits GTP-loaded mutant and WT KRAS, NRAS, HRAS.
- **n:** n=6-10/arm typical across studies
- **Effect size:** strong — Durable tumor regression in autochthonous PDAC GEMMs with KRAS G12D/V/R, including disease-stabilization in extensively metastatic mice; normal tissues showed transient proliferative arrest only, supporting a therapeutic window. CRC xenografts also regressed. The structural data underpins the case for activity in non-codon-12 RAS-pathway-activating alleles like A59T.
- **Variance:** vs vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PDAC-heavy package; CRC and atypical-KRAS data are subordinate. RMC-7977 is the tool compound; daraxonrasib (RMC-6236) is the clinical molecule the patient could access on trial.

Varkaris/Scaltriti (2024) — Cancer Discov

Reference:PMID 38078938 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Allosteric PI3K-alpha inhibition (RLY-2608) overcomes orthosteric resistance
- **Indication / model:** Engineered PIK3CA-mutant cell lines harboring secondary resistance mutations; xenograft validation
- **Design:** Allosteric (non-ATP-competitive) PI3K-alpha inhibition that binds a site distinct from kinase-domain orthosteric inhibitors. Retains activity against tumors that have acquired secondary resistance kinase-domain mutations and avoids hyperglycemia at active doses.
- **n:** isogenic panels + multiple PDX
- **Effect size:** strong — RLY-2608 retained activity in PIK3CA-mutant tumors that escaped alpelisib/inavolisib through secondary kinase-domain mutations, with comparable regression depth and improved metabolic tolerability. Relevant if the patient's M1043I or any acquired mutation impairs orthosteric binding.
- **Variance:** vs alpelisib- and inavolisib-resistant clones; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Allosteric agents are still trial-only. Useful as a downstream backup if first-generation PI3Kα inhibitors fail.

Waight/Wilson (2024) — Cancer Discov

Reference:PMID 39283994 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Botensilimab Fc-enhanced anti-CTLA-4 - preclinical mechanism
- **Indication / model:** Mouse surrogate of botensilimab (mFc-engineered anti-mouse CTLA-4); syngeneic models including poorly immunogenic and ICI-refractory tumors; FcgammaR-knockout mice
- **Design:** Fc-engineering increases affinity for activating FcgammaRs (especially FcgammaRIV/CD16a), which (i) enhances Treg depletion in tumors, (ii) promotes myeloid antigen-presenting cell activation, and (iii) potentiates T-cell priming - the constellation needed to convert immune-cold MSS tumors into responsive ones.
- **n:** n=8-12 mice/arm; multiple syngeneic models
- **Effect size:** strong — Surrogate botensilimab regressed multiple ICI-refractory syngeneic models that resisted

Fc-WT anti-CTLA-4 and anti-PD-1, with rejection of rechallenge. Provides the mechanistic basis for the clinical activity of botensilimab + balstilimab in MSS CRC where conventional ipilimumab/nivolumab fails.
- **Variance:** vs isotype Fc-wildtype anti-CTLA-4; anti-PD-1 alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mouse Fc-engineered surrogate, not the clinical human IgG1. Tumor models are syngeneic, not orthotopic peritoneal. Clinical CRC ORR in early reports was ~22%, consistent with the preclinical 'turns cold tumors warm' framing.

Rovers/de Hingh (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Perioperative systemic chemo (FOLFOX/FOLFOXIRI ± bev) + CRS-HIPEC
- **Indication / model:** Resectable colorectal peritoneal metastases (CAIRO6) (neoadj); —
- **Design:** 2/3
- **Effect size:** — — OS, PFS
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Revolution Medicines/Revolution Medicines (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** RMC-6236 (daraxonrasib) monotherapy
- **Indication / model:** Advanced solid tumors with RAS mutations (codons 12, 13, 61) (2L+); KRAS/NRAS/HRAS at codons 12, 13, or 61
- **Design:** 1/2
- **Effect size:** — — Safety, ORR, PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Jhaveri/Turner (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Inavolisib + palbociclib + fulvestrant
- **Indication / model:** HR+/HER2− PIK3CA-mutant metastatic breast cancer (INAVO120) (1L); PIK3CA mutation
- **Design:** 2/3
- **Effect size:** — — PFS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Raghav/Yoshino (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Trastuzumab deruxtecan (T-DXd) 5.4 mg/kg vs 6.4 mg/kg Q3W
- **Indication / model:** HER2-overexpressing metastatic CRC (DESTINY-CRC02) (2L+); HER2 IHC 3+ or IHC 2+/ISH+
- **Design:** 2
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Dialectic Therapeutics/Dialectic Therapeutics (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DT2216 monotherapy
- **Indication / model:** Relapsed/refractory solid tumors and hematologic malignancies (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, RP2D
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Kopetz/Tabernero (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Encorafenib + cetuximab ± mFOLFOX6
- **Indication / model:** BRAF V600E-mutant metastatic CRC, 1L (BREAKWATER) (1L); BRAF V600E
- **Design:** 3
- **Effect size:** — — PFS, ORR, OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Kim/Kim (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Lenvatinib + pembrolizumab (LEAP-017)
- **Indication / model:** Metastatic CRC (3L+, MSS/MSI-H mixed) (3L+); —
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Gritstone bio/Gritstone bio (2024) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GRT-C901 + GRT-R902 personalized neoantigen vaccine + atezolizumab + ipilimumab + fluoropyrimidine-leucovorin
- **Indication / model:** MSS metastatic colorectal cancer (1L maintenance) (1L); MSS
- **Design:** 2/3
- **Effect size:** — — PFS, OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Prager/Tabernero (2023) — N Engl J Med

Reference:PMID 37133585 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 3L backbone (TAS-102 + bev). Mechanism does not target the patient's features.

- **Intervention:** Trifluridine/tipiracil + bevacizumab (SUNLIGHT, 3L)
- **Notes:** Excluded: Standard-of-care 3L backbone (TAS-102 + bev). Mechanism does not target the patient's features. — Pivotal RCT: mOS 10.8 vs 7.5 mo (HR 0.61, p<0.001). Establishes TAS-102 + bev as 3L SoC. Audit-trail row.

Dasari/Eng (2023) — Lancet

Reference:PMID 37331369 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 3L backbone (fruquintinib, global confirmatory). Mechanism does not target the patient's features.

- **Intervention:** Fruquintinib (FRESCO-2, global 3L+)
- **Notes:** Excluded: Standard-of-care 3L backbone (fruquintinib, global confirmatory). Mechanism does not target the patient's features. — Global confirmatory: mOS 7.4 vs 4.8 mo (HR 0.66). Drove US approval Nov 2023. Audit-trail row.

Strickler/Bekaii-Saab (2023) — Lancet Oncol

Reference:PMID 37210019 · **Type:** clinical · **Inclusion:** excluded — MOUNTAINEER tucatinib + trastuzumab in HER2+ RAS-WT mCRC. Patient has KRAS A59T (atypical RAS-mutated), so the RAS-WT label criterion fails on protocol text — even if HER2 returns amplified on pending NGS, MOUNTAINEER-label use is not gated open. HER2 is not among the patient's nominated targetable features at intake.

- **Intervention:** Tucatinib + trastuzumab (MOUNTAINEER, HER2+ RAS-WT mCRC)
- **Notes:** Excluded: MOUNTAINEER tucatinib + trastuzumab in HER2+ RAS-WT mCRC. Patient has KRAS A59T (atypical RAS-mutated), so the RAS-WT label criterion fails on protocol text — even if HER2 returns amplified on pending NGS, MOUNTAINEER-label use is not gated open. HER2 is not among the patient's nominated targetable features at intake. — ORR 38%, mDoR 12.4 mo, mPFS 8.2 mo, mOS 24.1 mo. FDA accelerated approval Jan 2023. Audit-trail row; patient's KRAS A59T blocks the RAS-WT label gate.

Fakih/Kuboki (2023) — N Engl J Med

Reference:PMID 37870968 · **Type:** clinical · **Inclusion:** excluded — Sotorasib + panitumumab in KRAS G12C mCRC. Sotorasib is a G12C-specific covalent inhibitor binding the switch-II cysteine that KRAS A59T does not provide. The drug does not target A59T in any plausible mechanistic sense.

- **Intervention:** Sotorasib + panitumumab (CodeBreaK 300, KRAS G12C mCRC)
- **Notes:** Excluded: Sotorasib + panitumumab in KRAS G12C mCRC. Sotorasib is a G12C-specific covalent inhibitor binding the switch-II cysteine that KRAS A59T does not provide. The drug does not target A59T in any plausible mechanistic sense. — 960 mg arm: ORR 26%, mPFS 5.6 vs 2.2 mo SoC (HR 0.49). Anchor for the KRAS-G12C therapy class — patient's A59T is NOT a covered allele for any approved KRAS-targeted drug. Critical caveat to surface to the patient and family.

Yaeger/Klempner (2023) — N Engl J Med

Reference:PMID 36546659 · **Type:** clinical · **Inclusion:** excluded — Adagrasib ± cetuximab in KRAS G12C mCRC. KRAS A59T lacks the switch-II cysteine that the G12C covalent inhibitor needs. Not in mechanism scope for A59T.

- **Intervention:** Adagrasib ± cetuximab (KRYSTAL-1, KRAS G12C mCRC)
- **Notes:** Excluded: Adagrasib ± cetuximab in KRAS G12C mCRC. KRAS A59T lacks the switch-II cysteine that the G12C covalent inhibitor needs. Not in mechanism scope for A59T. — Monotherapy ORR 19%; adagrasib + cetuximab ORR 46%, mDoR 7.6 mo. Drove FDA accelerated approval of the combination (Jun 2024) in G12C mCRC. A59T patient is not eligible.

Yamazaki/Melisi (2023) — BMC Cancer

Reference:PMID 37507657 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Galunisertib (TGFbR1 SMI) + nivolumab phase 1b/2 in solid tumors and NSCLC
- **Indication / model:** Advanced refractory solid tumors and recurrent NSCLC, hepatocellular carcinoma (2L+); n=75 across phase 1b dose-escalation and phase 2 expansion. Mixed tumor types — solid-tumor cohort included CRC.
- **Design:** Phase 1b/2 open-label dose-escalation and expansion
- **n:** 75
- **Effect size:** 5 % — Safety, RP2D, ORR
- **Variance:** PRs in NSCLC and HCC cohorts; CRC subset without confirmed responses
- **Toxicities:** Fatigue (40%); Diarrhea (25%); Cardiovascular AE (TGFb class effect) G>=3 (4%)
- **Case fit:** weak
- **Notes:** Closest TGFb-axis + ICI clinical readout. No CRC-specific responder signal; SMAD4-loss-stratified analysis not performed. Galunisertib program subsequently wound down at Lilly. Cross-tumor proof of tolerability only.

Vacca/Friedlander (2023) — Mol Cancer Ther (AACR-NCI-EORTC poster)

Reference:doi:10.1158/1535-7163.TARG-23-B114 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ST316 peptide antagonist of beta-catenin/BCL9
- **Indication / model:** CRC cell-line panel and xenografts including APC-mutant lines; in vivo combination with anti-PD-1
- **Design:** Cell-penetrating peptide that disrupts beta-catenin/BCL9 interaction, blocking nuclear beta-catenin

transcriptional activity downstream of APC loss. Acts at the transcription complex rather than upstream of ligand secretion, making it potentially active in APC-mutant tumors that escape PORCN inhibition.
- **n:** n=8-10/arm typical
- **Effect size:** moderate — ST316 suppressed CRC xenograft growth, including APC-mutant models, and synergized with anti-PD-1 in syngeneic tumors. Mechanism plausibly engages the patient's APC-E1295-driven Wnt activation in a way PORCN inhibitors cannot.
- **Variance:** vs vehicle and isotype control; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Preclinical data are mostly conference abstracts; full peer-reviewed mechanism + efficacy paper not yet published. Phase 2 monotherapy + combination CRC trial is ongoing.

Carozza/Li (2023) — Cancer Res (AACR abstract LB323)

Reference:doi:10.1158/1538-7445.AM2023-LB323 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SR-8541A ENPP1 inhibitor - STING-axis amplification in CRC
- **Indication / model:** CT26 mouse CRC; in vitro PBMC IFN-beta induction; combination with anti-PD-1 or anti-CTLA-4
- **Design:** ENPP1 hydrolyzes the STING agonist 2'3'-cGAMP in the tumor microenvironment, dampening innate immune activation. ENPP1 inhibition raises extracellular cGAMP, activates STING in DCs and macrophages, and increases type-I IFN signaling - a complement to Fc-enhanced CTLA-4 and PD-1 blockade in cold tumors.
- **n:** n=8-12 mice/arm
- **Effect size:** moderate — SR-8541A produced CT26 tumor regression as monotherapy and synergized with ICI to extend survival. Provides the rationale for combining ENPP1 inhibition with botensilimab/balstilimab in MSS CRC, including the trial arm the patient's case lists.
- **Variance:** vs vehicle, ICI alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Conference-abstract data; full peer-reviewed publication pending. CT26 is the standard MSS CRC model but tumor immunogenicity may exceed real MSS CRC.

Korean Cancer Study Group/Korean Cancer Study Group (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Alpelisib + capecitabine
- **Indication / model:** PIK3CA-mutant metastatic colorectal cancer (2L+); PIK3CA mutation
- **Design:** 1/2
- **Effect size:** — — Safety, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Strickler/Bekaii-Saab (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tucatinib + trastuzumab
- **Indication / model:** HER2+ metastatic colorectal cancer (MOUNTAINEER) (2L+); HER2+ (IHC 3+ or IHC 2+/ISH amplified, or NGS amplification) AND RAS-WT
- **Design:** 2
- **Effect size:** — — ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Aprea/Aprea (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) + pembrolizumab
- **Indication / model:** Solid tumors with TP53 mutation (bladder, gastric, NSCLC) (2L+); TP53 mutation
- **Design:** 1/2
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Fang/Fang (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** GUCY2C CAR-T cells
- **Indication / model:** GUCY2C-positive digestive tract tumors (2L+); GUCY2C-positive
- **Design:** early phase 1
- **Effect size:** — — Safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Prager/Tabernero (2023) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Trifluridine/tipiracil + bevacizumab
- **Indication / model:** Refractory metastatic colorectal cancer (SUNLIGHT) (3L+); —
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Park/Park (2022) — ESMO Open

Reference:PMID 36018990 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt + pembrolizumab phase 1b in TP53-mutant advanced solid tumors
- **Indication / model:** Advanced solid tumors with TP53 mutation (2L+); n=24, mixed TP53-mut solid tumors (NSCLC, urothelial, gastric, prostate, others). TP53 mutations spanned hotspots and non-hotspots; specific R273 subset not separately analyzed.
- **Design:** Phase 1b single-arm
- **n:** 24
- **Effect size:** 8 % — Safety and ORR
- **Variance:** 2 PRs in 24 evaluable
- **Toxicities:** Fatigue (50%); Ataxia/encephalopathy G>=3 (12%)
- **Case fit:** weak
- **Notes:** Modest single-arm signal in TP53-mut solid tumors; not enough to support routine use in TP53 R273 mCRC. The Aprea program effectively closed after the company pivoted away from solid tumors. R273-specific pharmacology is not addressed.

Yi/Yi (2022) — Clin Cancer Res / ASCO 2022

Reference: — · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pelcitoclax (APG-1252) dual Bcl-2/Bcl-xL inhibitor phase 1
- **Indication / model:** SCLC and other solid tumors (2L+); Mixed solid tumors, SCLC-enriched cohort.
- **Design:** Phase 1 single-arm dose-escalation
- **n:** 39
- **Effect size:** modest activity — — Safety, RP2D, anti-tumor activity
- **Variance:** few PRs; thrombocytopenia dose-limiting
- **Toxicities:** Thrombocytopenia G>=3 (30%)
- **Case fit:** weak
- **Notes:** Limited published data; Ascentage program has decelerated post-2023. Audit-trail row for BCL-xL class. AE detail incomplete in available abstracts.

Song/Friedman (2022) — J Med Chem

Reference:PMID 36455032 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Inavolisib (GDC-0077) selective inhibitor + degrader of mutant p110-alpha
- **Indication / model:** PIK3CA-mutant breast cancer cell lines and xenografts (KPL-4, HCC1954, MCF-7), some CRC lines
- **Design:** ATP-competitive PI3K-alpha inhibitor with secondary HER2-dependent induced degradation of mutant p110-alpha. Achieves deeper, longer pathway suppression than orthosteric-only alpelisib in PIK3CA-mutant cells.
- **n:** n=8-10 mice/arm typical
- **Effect size:** strong — Inavolisib regressed PIK3CA-mutant breast xenografts at lower exposures than alpelisib, with sustained pSer473-AKT suppression and selective degradation of mutant (not WT) p110-alpha in HER2-high contexts. Activity in M1043 kinase-domain mutants matches the helical-mutant pattern in side-by-side panels.
- **Variance:** vs vehicle; alpelisib and taselisib comparators; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Most in vivo data in HER2+ or ER+ breast models; CRC-specific PDX data are limited. HER2 status will

modulate the degrader arm of mechanism in this patient (HER2 not yet reported).

Ghosh/Maecker (2022) — Sci Transl Med

Reference:PMID 36127498 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** APR-246 + anti-PD-1 combination in solid tumors
- **Indication / model:** Syngeneic mouse tumor models with TP53 mutations; co-treatment with anti-PD-1
- **Design:** Pharmacologically increased p53 signaling reprograms tumor-associated macrophages toward M1 and disinhibits antitumor T cells, augmenting anti-PD-1 efficacy.
- **n:** n=8-10 mice/arm typical
- **Effect size:** moderate — Combination produced sustained tumor regression where either agent alone showed modest activity. Mechanistic basis for the eprenetapopt + pembrolizumab clinical combination strategy in TP53-mutant tumors.
- **Variance:** vs vehicle, APR-246 alone, anti-PD-1 alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Syngeneic models, not orthotopic CRC; eprenetapopt phase 1/2 in solid tumors with pembrolizumab has so far underperformed the preclinical signal. Useful background but not a stand-alone case for off-trial use in this MSS patient.

Yi/Zhou (2022) — J Hematol Oncol

Reference:PMID 35260176 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DT2216 + KRAS G12C inhibitor (sotorasib) synergy
- **Indication / model:** KRAS-G12C-mutant NSCLC and CRC cell lines and xenografts
- **Design:** BCL-xL upregulation is a key resistance node to KRAS-G12C inhibition. BCL-xL degradation re-sensitizes cells to apoptosis triggered by KRAS pathway suppression, including in CRC where G12C monotherapy underperforms NSCLC.
- **n:** n=8-10/arm xenograft
- **Effect size:** strong — DT2216 + sotorasib gave durable regressions in KRAS-G12C CRC xenografts where sotorasib alone showed regrowth. Demonstrates the BCL-xL-as-apoptotic-priming concept transferable to other RAS-pathway interventions (potentially pan-RAS RMC-6236 in the patient's A59T setting).
- **Variance:** vs sotorasib alone; DT2216 alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Direct read-across to A59T + pan-RAS inhibitor is inference; no dedicated A59T model published. Combination strategy is the most translatable insight.

Yonesaka/Nakagawa (2022) — Jpn J Clin Oncol

Reference:PMID 35671377 · **Type:** preclinical · **Inclusion:** excluded — ADC delivering a TOP1-inhibitor payload (deruxtecan). Patient is TOP1-amplified at 20q11 but HER2 status not reported. Promote to 'included' if HER2 IHC 2+/3+ or ISH-positive. Conceptually interesting because TOP1 amplification could amplify deruxtecan-payload sensitivity even at modest HER2 expression.

- **Intervention:** Trastuzumab deruxtecan HER2-low CRC PDX preclinical activity

- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: ADC delivering a TOP1-inhibitor payload (deruxtecan). Patient is TOP1-amplified at 20q11 but HER2 status not reported. Promote to 'included' if HER2 IHC 2+/3+ or ISH-positive. Conceptually interesting because TOP1 amplification could amplify deruxtecan-payload sensitivity even at modest HER2 expression.

Ascentage Pharma/Ascentage Pharma (2022) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Pelcitoclax (APG-1252) monotherapy
- **Indication / model:** SCLC and other solid tumors (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, RP2D
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Quenet/Glehen (2021) — Lancet Oncol

Reference:PMID 33417845 · **Type:** clinical · **Inclusion:** excluded — Surgical / locoregional intervention (CRS-HIPEC vs CRS alone). Not a feature-targeting drug. Surfaced separately as a critical sequencing context for the planned HIPEC window; the treating team owns this decision.

- **Intervention:** Oxaliplatin HIPEC + CRS vs CRS alone (PRODIGE 7)
- **Notes:** Excluded: Surgical / locoregional intervention (CRS-HIPEC vs CRS alone). Not a feature-targeting drug. Surfaced separately as a critical sequencing context for the planned HIPEC window; the treating team owns this decision. — Critical negative trial: oxaliplatin HIPEC added to CRS did NOT improve OS (41.7 vs 41.2 mo) or RFS (13.1 vs 11.1 mo) in CRC peritoneal metastases, with higher 60-day morbidity in the HIPEC arm. CRS retains the OS signal; HIPEC does not. This is the central caveat to flag for the patient's planned peri-operative HIPEC window.

Rovers/de Hingh (2021) — Ann Surg Oncol

Reference:PMID 33544279 · **Type:** clinical · **Inclusion:** excluded — PIPAC monotherapy in unresectable CPM. Locoregional drug delivery, not feature-targeted. Mechanism rationale (oxaliplatin via aerosol) does not address KRAS A59T, PIK3CA M1043I, or other features. Audit-trail row.

- **Intervention:** Oxaliplatin PIPAC (CRC-PIPAC, unresectable CPM)
- **Notes:** Excluded: PIPAC monotherapy in unresectable CPM. Locoregional drug delivery, not feature-targeted. Mechanism rationale (oxaliplatin via aerosol) does not address KRAS A59T, PIK3CA M1043I, or other features. Audit-trail row. — Phase 2 single-arm, n=20: mPFS 3.5 mo, mOS 8.0 mo, 15% major AE rate. Decision-relevant if upfront CRS-HIPEC is deferred (trial row NCT04329494 maps to City of Hope PIPAC platform).

Siena/Yoshino (2021) — Lancet Oncol

Reference:PMID 33872572 · **Type:** clinical · **Inclusion:** excluded — HER2+ ADC (T-DXd). HER2 is NOT among the patient's nominated targetable features; HER2 status is not reported on profile.json. Out of Libby's feature scope until/unless HER2 amp returns on pending NGS.

- **Intervention:** Trastuzumab deruxtecan (DESTINY-CRC01, HER2-expressing mCRC)
- **Notes:** Excluded: HER2+ ADC (T-DXd). HER2 is NOT among the patient's nominated targetable features; HER2 status is not reported on profile.json. Out of Libby's feature scope until/unless HER2 amp returns on pending NGS. — Cohort A (HER2 IHC 3+ or 2+/ISH+): ORR 45.3%, mPFS 6.9 mo. Cross-tumor / cross-biomarker reference. Audit-trail row.

Arena/Siena (2021) — Clin Cancer Res

Reference:PMID 34031055 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Functional / clinical characterization of atypical KRAS/NRAS mutations in mCRC
- **Indication / model:** Metastatic colorectal cancer with atypical (non-G12/G13/Q61) KRAS or NRAS mutations (any); Cohort of mCRC patients with atypical KRAS/NRAS alterations — A59, K117, A146, others — profiled functionally and clinically; subset received anti-EGFR therapy.
- **Design:** Retrospective cohort with functional characterization
- **n:** 169
- **Effect size:** variable by allele % — Anti-EGFR response rate and PFS in atypical-KRAS subset
- **Variance:** A146/A59 alleles show partial residual sensitivity in some cases; functional assays show weaker MAPK output than G12 mutants
- **Toxicities:** Not assessed.
- **Case fit:** strong
- **Notes:** Most directly relevant atypical-KRAS / anti-EGFR paper for the case — functional and clinical heterogeneity at A59 and adjacent codons. PMC8364867. Supports the position that A59T should be assessed individually rather than treated as a flat anti-EGFR contraindication.

Vidal/Beg (2021) — Oncologist / Mol Cancer Res

Reference:PMID 33945173 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Immuno-oncology biomarker enrichment in KRAS A59T tumors
- **Indication / model:** Solid tumors (colorectal predominant) harboring KRAS A59T (any); Caris real-world cohort: KRAS A59T variant cases compared to KRAS WT and other KRAS mutants for PD-L1, TMB, MSI, and immune-microenvironment markers.
- **Design:** Retrospective real-world biomarker cohort
- **n:** 280
- **Effect size:** enriched descriptive — Enrichment of immuno-oncology biomarkers in A59T tumors
- **Variance:** PD-L1 + TMB higher in A59T-mut vs other KRAS-mut CRC
- **Toxicities:** Not applicable (observational).
- **Case fit:** strong
- **Notes:** The patient is MSS / TMB-low / PD-L1-neg by intake — the opposite of the A59T-enriched pattern described here. The A59T immuno-oncology hypothesis does not light up for this individual. Worth surfacing so the board knows the IO-enrichment signal exists at the allele level but doesn't apply here. PMC8224072.

Rodon/Cheng (2021) — Br J Cancer

Reference:PMID 33875737 · **Type:** clinical · **Inclusion:** included

- **Intervention:** WNT974 (LGK974) porcupine inhibitor phase 1 in advanced solid tumors

- **Indication / model:** Advanced solid tumors, biomarker-unselected (Wnt pathway hypothesis) (2L+); n=94 advanced solid tumors. AXIN2 expression by RT-PCR used as PD biomarker.
- **Design:** Phase 1 dose-escalation, single-arm
- **n:** 94
- **Effect size:** 16 % — Safety, RP2D, target engagement (AXIN2 reduction), ORR
- **Variance:** no ORs by RECIST 1.1; SD 16%, mDoR ~20 wks
- **Toxicities:** Dysgeusia (50%); Decreased appetite (30%); Fatigue (28%); Fragility fractures / osteopenia (10%)
- **Case fit:** weak
- **Notes:** Foundational porcupine-inhibitor readout — pharmacologically active on Wnt but no objective responses, supporting the field-wide skepticism that 'just inhibit Wnt secretion' is insufficient for APC-mutant CRC. APC-loss tumors are RNF43-fusion-naive and likely resistant to porcupine inhibition. Audit-trail row anchoring the Wnt-class evidence base.

Arena/Bardelli (2021) — Clin Cancer Res

Reference:PMID 34117033 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Atypical KRAS/NRAS variant functional characterization (incl. A59 codon)
- **Indication / model:** Ba/F3 IL-3-independent transformation assay; isogenic NIH-3T3 transduction; in vivo xenografts of transduced clones; correlated against 9,485-patient tissue + ctDNA cohorts
- **Design:** Functional annotation of 57 non-canonical KRAS/NRAS alleles by Ba/F3 IL-3 withdrawal, ERK/AKT signaling, and xenograft growth. A59-class variants (A59E/A59T/A59G) sit in the switch-II region and reduce intrinsic GTP hydrolysis, but with weaker MAPK output than G12/Q61 controls in some clones.
- **n:** 57 atypical variants tested; 9,485 patient samples
- **Effect size:** moderate — About a third of atypical variants signaled below WT, 40% sat between WT and G12V, and 28% were hyperactive. A59 mutants stratified as activating but variable; the paper does not single out A59T with its own xenograft regression curve, so functional class assignment is the load-bearing translatable result for anti-EGFR exclusion decisions.
- **Variance:** vs KRAS WT and KRAS G12V/G12D activating controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Ba/F3 + xenograft readouts are surrogate for clinical anti-EGFR resistance; A59T-specific tumor regression data are sparse. Functional class call drives the decision; the paper is the standard reference for treating A59T as RAS-activating.

Lou/Korn (2021) — Cells

Reference:PMID 34063999 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** KRAS A59T variant co-mutation and immuno-oncologic biomarker landscape
- **Indication / model:** Caris real-world genomic + IHC database (17,909 CRC + pan-tumor cases); no in-vivo arm
- **Design:** Profiling of A59T-mutant tumors for MSI-H/dMMR, high TMB, and PD-L1 positivity relative to A59T-WT CRC.
- **n:** 14 A59T-mutant CRC cases (0.08% prevalence); 19 pan-tumor A59T cases
- **Effect size:** moderate — A59T-mutant CRC was strongly enriched for MSI-H/dMMR (~50%), high TMB and PD-L1 positivity. The patient here is MSS/TMB-low/PD-L1-neg, putting them in the minority A59T subset that does not inherit that immune-active phenotype; this argues against generalizing ICI-favorable A59T literature to this case.

- **Variance:** vs Non-A59T KRAS-mutant and KRAS WT CRC; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Descriptive cohort study, not mechanistic. Useful for caveating immune-axis assumptions about A59T tumors when the index case is MSS.

Degtjarik/Shakked (2021) — Nat Commun

Reference:PMID 29618700 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** APR-246 (eprenetapopt) covalent reactivator of mutant p53
- **Indication / model:** Biochemical thermostability + crystal structure + cell-line panels including R175H, R273H, R248Q mutants
- **Design:** APR-246 is converted to methylene-quinuclidinone (MQ), which forms covalent adducts at Cys277 (and Cys124 for R175H), thermostabilizing the p53 fold and restoring WT-like transactivation; additionally depletes glutathione and inhibits TrxR1, inducing ROS.
- **n:** biochemical replicates n=3+; structural data
- **Effect size:** moderate — Cys277 is required for MQ-mediated thermostabilization of R273H mutant p53, providing direct biophysical evidence for R273H reactivation. The dual ROS/p53-reactivation mechanism has driven multiple preclinical CRC and ovarian xenograft regressions in subsequent literature.
- **Variance:** vs WT p53 and p53-null; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Strong preclinical mechanism, but clinical translation failed in the pivotal MDS phase 3 trial. Background literature only - not a basis for repurposing without trial context.

Chen/Lu (2021) — Cancer Cell

Reference:PMID 33357454 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Arsenic trioxide rescues structural p53 mutants via cryptic allosteric site
- **Indication / model:** p53-mutant cell-line panel (R175H, Y220C, R249S, structural-class); biochemical and crystallographic structures of arsenic-bound p53; xenograft series
- **Design:** Arsenic coordinates three cysteines in a cryptic allosteric pocket of the p53 DNA-binding domain, restoring fold integrity of structural-class mutants. Distinct mechanism from APR-246 (electrophile at Cys277).
- **n:** structural studies + xenograft n=6-8/arm
- **Effect size:** moderate — Structural-class mutants (R175H, Y220C, R249S) showed restored DNA binding and tumor regression; contact-class R273H was less responsive in the original panel. Frames a parallel-track mechanism but argues against ATO as a primary play for this patient's R273 (DNA-contact) hotspot until the substitution (R273H vs R273C vs R273L) is confirmed.
- **Variance:** vs vehicle, WT p53 lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Contact-class R273H sits outside the strongest activity tier in the Chen paper. ATO is included for mechanism context and as a 'don't-extrapolate-blindly' reference, not as a pursued lead.

Sjoblom/Schweiger (2021) — Mol Cancer Res

Reference:PMID 33619228 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SMAD4 R361 hotspot mutations boost Wnt/beta-catenin signaling via LEF1
- **Indication / model:** Engineered SMAD4-R361 mutant CRC cell lines; pull-down + reporter assays; xenograft growth
- **Design:** R361 hotspot mutants in the MH2 domain disrupt heterodimer formation with SMAD2/3 (canonical TGF-beta loss-of-function) while simultaneously gaining enhanced binding to LEF1, amplifying Wnt/beta-catenin transcription. A bona fide gain-of-function loss-of-function dual phenotype.
- **n:** n=3 biological replicates; xenograft n=6/arm
- **Effect size:** strong — SMAD4 R361 mutants showed increased Wnt reporter activity and accelerated xenograft growth versus null. Means the patient's SMAD4 R361H is not simply 'TGF-beta loss' - it actively amplifies the Wnt axis driven by APC E1295, with implications for Wnt-targeted strategies and for the case's CMS4-mesenchymal lean.
- **Variance:** vs WT SMAD4 isogenic; SMAD4-null; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line and xenograft work, no GEMM. The dual mechanism is unusual and worth flagging when discussing Wnt-pathway and TGF-beta-axis interventions together.

Cremolini/Falcone (2020) — Lancet Oncol

Reference:PMID 32007158 · **Type:** clinical · **Inclusion:** excluded — Backbone sequencing trial (FOLFOXIRI-bev upfront + reintroduction vs sequential FOLFOX-bev then FOLFIRI-bev). Mechanism does not target the patient's features.

- **Intervention:** Upfront FOLFOXIRI + bev (TRIBE2)
- **Notes:** Excluded: Backbone sequencing trial (FOLFOXIRI-bev upfront + reintroduction vs sequential FOLFOX-bev then FOLFIRI-bev). Mechanism does not target the patient's features. — Strategy trial — upfront triplet improved mPFS2 (19.2 vs 16.4 mo, HR 0.74) and mOS (27.4 vs 22.5 mo, HR 0.82). Audit-trail row.

Goéré/Elias (2020) — Lancet Oncol

Reference:PMID 32717181 · **Type:** clinical · **Inclusion:** excluded — Second-look surgery + HIPEC for patients at high risk of peritoneal recurrence (PROPHYLOCHIP/PRODIGE 15). Surgical intervention; not feature-targeting.

- **Intervention:** Second-look surgery + HIPEC vs surveillance (PROPHYLOCHIP-PRODIGE 15)
- **Notes:** Excluded: Second-look surgery + HIPEC for patients at high risk of peritoneal recurrence (PROPHYLOCHIP/PRODIGE 15). Surgical intervention; not feature-targeting. — 3-yr DFS 44% vs 53% (HR 0.97, p=0.82). No benefit from prophylactic second-look + HIPEC. Audit-trail row reinforcing HIPEC skepticism for the planned window.

Fukuoka/Doi (2020) — J Clin Oncol

Reference:PMID 32167863 · **Type:** clinical · **Inclusion:** excluded — REGONIVO regorafenib + nivolumab in MSS gastric/CRC. Anti-angiogenic + anti-PD-1 backbone — mechanism does not target the patient's nominated features. Listed in audit trail as the foundational MSS-CRC ICI-combination signal (later largely failed to replicate in Western cohorts).

- **Intervention:** Regorafenib + nivolumab (REGONIVO, EPOC1603) MSS gastric/CRC
- **Notes:** Excluded: REGONIVO regorafenib + nivolumab in MSS gastric/CRC. Anti-angiogenic + anti-PD-1 backbone — mechanism does not target the patient's nominated features. Listed in audit trail as the foundational MSS-CRC ICI-combination signal (later largely failed to replicate in Western cohorts). — Japanese phase 1b: ORR 36% in MSS CRC (n=25). Western replication studies (Fakih 2021, eClinicalMedicine 2023) failed to confirm. Audit-trail row for MSS-CRC ICI landscape.

Kulukian/Olson (2020) — Mol Cancer Ther

Reference:PMID 32430484 · **Type:** preclinical · **Inclusion:** excluded — HER2-selective TKI; patient HER2 status not yet reported. If HER2-amplified on confirmation, this row should be promoted to 'included' with the Kulukian 2020 PDX panel data (12 PDX + 2 CDX showing 85-137% combination tumor growth inhibition with trastuzumab). Listed in audit trail pending HER2 IHC/ISH.

- **Intervention:** Tucatinib + trastuzumab HER2+ CRC PDX panel (Kulukian)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: HER2-selective TKI; patient HER2 status not yet reported. If HER2-amplified on confirmation, this row should be promoted to 'included' with the Kulukian 2020 PDX panel data (12 PDX + 2 CDX showing 85-137% combination tumor growth inhibition with trastuzumab). Listed in audit trail pending HER2 IHC/ISH.

Parikh/Lenz (2019) — Clin Cancer Res

Reference:PMID 30224341 · **Type:** clinical · **Inclusion:** excluded — Backbone comparison (FOLFIRI-bev vs FOLFOX-bev stratified by ERCC1). Mechanism does not target the patient's features.

- **Intervention:** mFOLFOX6-bev vs FOLFIRI-bev (MAVERICC, biomarker-stratified)
- **Notes:** Excluded: Backbone comparison (FOLFIRI-bev vs FOLFOX-bev stratified by ERCC1). Mechanism does not target the patient's features. — Phase 2 biomarker-stratified RCT: PFS/OS comparable between arms; numeric trend favored FOLFIRI-bev. Anchors the 1L FOLFIRI-bev choice the patient is already on.

Bekaii-Saab/Goldberg (2019) — Lancet Oncol

Reference:PMID 31262657 · **Type:** clinical · **Inclusion:** excluded — Regorafenib dose-optimization (ReDOS, 80→160 mg ramp). Mechanism does not target the patient's features; useful tolerance-management evidence for the treating team.

- **Intervention:** Regorafenib dose-escalation (ReDOS, 80→120→160 mg)
- **Notes:** Excluded: Regorafenib dose-optimization (ReDOS, 80→160 mg ramp). Mechanism does not target the patient's features; useful tolerance-management evidence for the treating team. — Ramp-up dosing matched the standard 160 mg arm on activity with lower G3 AE incidence (hand-foot, fatigue, HTN, diarrhea). Audit-trail row.

Klaver/Tanis (2019) — Lancet Gastroenterol Hepatol

Reference:PMID 31272834 · **Type:** clinical · **Inclusion:** excluded — Adjuvant oxaliplatin HIPEC after primary resection for high-risk T4/perforated colon cancer. Surgical locoregional intervention; not a feature-targeting drug.

- **Intervention:** Adjuvant oxaliplatin HIPEC for T4/perforated colon cancer (COLOPEC)

- **Notes:** Excluded: Adjuvant oxaliplatin HIPEC after primary resection for high-risk T4/perforated colon cancer. Surgical locoregional intervention; not a feature-targeting drug. — Adjuvant HIPEC did NOT improve 18-mo peritoneal-disease-free survival (80.9% vs 76.2%). Together with PRODIGE 7 and PROPHYLOCHIP, completes the negative HIPEC trifecta in colon cancer.

Eng/Bendell (2019) — Lancet Oncol

Reference:PMID 30975633 · **Type:** clinical · **Inclusion:** excluded — Atezolizumab ± cobimetinib vs regorafenib in 3L+ mCRC (mostly MSS). Negative phase 3 — ICI + MEK inhibition does not target the patient's nominated features.

- **Intervention:** Atezolizumab ± cobimetinib vs regorafenib (IMblaze370, 3L+ mCRC)
- **Notes:** Excluded: Atezolizumab ± cobimetinib vs regorafenib in 3L+ mCRC (mostly MSS). Negative phase 3 — ICI + MEK inhibition does not target the patient's nominated features. — Negative: mOS 8.87 (atezo+cobi) vs 7.10 (atezo) vs 8.51 (regorafenib) mo. Reinforces that MAPK + PD-L1 blockade is insufficient for immune-excluded MSS CRC.

Vasan/Scaltriti (2019) — Science

Reference:PMID 31699932 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Double cis-PIK3CA mutations sensitize to PI3K-alpha inhibition
- **Indication / model:** Isogenic MCF10A, breast and colorectal cancer cell lines, PDX, retrospective clinical cohort
- **Design:** Two cis PIK3CA activating mutations (kinase + helical or two kinase) hyperactivate p110-alpha beyond a single mutation, increasing tumor dependence on PI3Kα and amplifying response to PI3K-alpha-selective inhibition.
- **n:** isogenic panels + PDX + ~3,000 patient samples reanalyzed
- **Effect size:** strong — Double cis PIK3CA mutants showed higher PIP3 production, more transformation, and significantly deeper xenograft regression on alpelisib/taselisib than single mutants. M1043 is one of the recurrent kinase-domain hits cited in the double-mutant series, supporting the index case's M1043I as PI3Kα-inhibitor-sensitive biology even at single-allele dosing.
- **Variance:** vs single-mutant PIK3CA and WT; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Breast-cancer-heavy in vivo work. CRC translatability has been weaker historically (KRAS co-mutation blunts alpelisib monotherapy), so synergy with anti-RAS or chemo is the practical play.

Khan/Zhou (2019) — Nat Med

Reference:PMID 31792461 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** DT2216 BCL-xL PROTAC degrader - platelet-sparing mechanism
- **Indication / model:** BCL-xL-dependent leukemia (MOLT-4) and solid-tumor (small-cell lung, breast) xenografts; primary platelets ex vivo
- **Design:** PROTAC bifunctional molecule recruits VHL E3 ligase to BCL-xL, triggering its ubiquitination and proteasomal degradation. Platelets express very low VHL, so they are largely spared - decoupling BCL-xL antitumor activity from on-target thrombocytopenia.
- **n:** xenograft n=8-10/arm; platelet n=3 donors

- **Effect size:** strong — DT2216 was more potent than navitoclax against BCL-xL-dependent xenografts at doses that caused minimal platelet drop in mice. Solves the navitoclax dose-limiting toxicity problem on paper; supports DT2216 as the preferred BCL-xL strategy for the patient's BCL2L1-amplified tumor, particularly in combination with irinotecan given the TOP1 co-amplification.
- **Variance:** vs ABT-263 (navitoclax) head-to-head; vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Solid-tumor in vivo data are mostly SCLC and breast; CRC-specific in vivo combination data with irinotecan is in conference-abstract form. Phase 1 in solid tumors is now reporting and validates the platelet-sparing claim.

Li/Xu (2018) — JAMA

Reference:PMID 29946728 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 3L backbone (fruquintinib). Anti-VEGFR1/2/3 mechanism does not target the patient's features.

- **Intervention:** Fruquintinib (FRESCO, Chinese 3L+)
- **Notes:** Excluded: Standard-of-care 3L backbone (fruquintinib). Anti-VEGFR1/2/3 mechanism does not target the patient's features. — Chinese pivotal: mOS 9.3 vs 6.6 mo (HR 0.65). Audit-trail row.

Lochhead/Lochhead (2018) — Case Rep Oncol

Reference:PMID 30574100 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Panitumumab + FOLFIRI in KRAS A59T mCRC (case report)
- **Indication / model:** Metastatic colon cancer harboring KRAS A59T (2L+); Single patient with KRAS A59T (initially read as KRAS exon-2 WT on limited testing; A59T disclosed on broader NGS after FOLFIRI-bev progression). Switched to panitumumab + FOLFIRI on disease progression.
- **Design:** Single-patient case report
- **n:** 1
- **Effect size:** PR (~36% reduction) % — Radiographic PR (~36% reduction) with concurrent CEA decline; SD continued.
- **Variance:** single patient — descriptive only
- **Toxicities:** No new safety signals; standard panitumumab + FOLFIRI tolerance.
- **Case fit:** strong
- **Notes:** Decision-relevant precedent — challenges blanket assumption that all atypical KRAS mutations confer anti-EGFR resistance. A single PR in a 37yo with A59T is hypothesis-generating, not confirmatory; balance against Sorich/Schirripa meta-analytic signal that non-canonical RAS mutations attenuate anti-EGFR benefit. PMC6295659. AE detail not captured beyond 'no new safety signals' — abstract-only in PMC, no AE table.

Juric/Baselga (2018) — J Clin Oncol

Reference:PMID 29401002 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Alpelisib (BYL719) monotherapy in PIK3CA-altered solid tumors (first-in-human)
- **Indication / model:** Advanced solid tumors with PIK3CA alterations (mixed: breast, CRC, gynecologic, endometrial) (2L+); n=134 PIK3CA-altered solid tumors; CRC subset included. PIK3CA mutations spanned the canonical helical / kinase hotspots (E542K, E545K, H1047R) and rarer alleles.

- **Design:** Phase 1a dose-escalation and expansion, single-arm
- **n:** 134
- **Effect size:** 6 % — ORR and clinical activity across tumor types
- **Variance:** 8/134 PRs total (1 breast, 2 CRC, 4 gyn/endometrial, 1 other)
- **Toxicities:** Hyperglycemia G>=3 (20%); Rash/maculopapular G>=3 (10%); Diarrhea G>=3 (7%); Nausea (56%)
- **Case fit:** partial
- **Notes:** Anchor for alpelisib's PIK3CA-mutant pharmacology in solid tumors — including the 2 CRC responders. Closest standalone CRC + PIK3CA-targeting clinical readout outside breast cancer.

Overman/André (2018) — J Clin Oncol

Reference:PMID 29355075 · **Type:** clinical · **Inclusion:** excluded — Nivolumab + ipilimumab in dMMR/MSI-H mCRC (CheckMate 142). Patient is MSS — out of biomarker scope.

- **Intervention:** Nivolumab + ipilimumab (CheckMate 142, MSI-H mCRC)
- **Notes:** Excluded: Nivolumab + ipilimumab in dMMR/MSI-H mCRC (CheckMate 142). Patient is MSS — out of biomarker scope. — MSI-H only; ORR 55% (4-yr update 65%). Audit-trail row defining the MSI-H exception.

Nichols/Shokat (2018) — Nat Cell Biol

Reference:PMID 30104724 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SHP2 inhibition (SHP099/RMC-4630/TNO155) in nucleotide-cycling RAS-pathway tumors
- **Indication / model:** KRAS-mutant NSCLC and CRC cell lines; PDX; BRAF class 3 and NF1-LOF models
- **Design:** SHP2 (PTPN11) acts upstream of RAS to load it with GTP via SOS1; inhibition is most effective when KRAS retains intrinsic GTP hydrolysis (nucleotide-cycling alleles such as G12C, G13D, atypical alleles with preserved cycling).
- **n:** panels of 10-20+ lines
- **Effect size:** moderate — Allosteric SHP2 inhibitors blocked growth of tumors driven by 'semi-autonomous' RAS-pathway alleles - G12C, NF1-LOF, BRAF class 3 - by interrupting RTK-to-RAS-GTP loading. A59 mutants partially retain GTPase activity, which is the rationale (not direct in-vivo evidence) for SHP2 sensitivity here.
- **Variance:** vs vehicle and RAS-pathway-WT controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Q61 and G12V/D alleles that disable GTPase activity are largely SHP2-insensitive; A59 sits in switch-II and its GTPase status is allele-specific. Use this only to frame the rationale, not as direct A59T evidence.

Tauriello/Batlle (2018) — Nature

Reference:PMID 29443960 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TGF-beta blockade + anti-PD-L1 in genetically reconstituted CRC metastasis
- **Indication / model:** Quadruple-mutant (Apc, Kras, Tgfbr2, Trp53) mouse CRC organoid; orthotopic and liver-metastasis syngeneic transplant; PD-L1 + galunisertib combination
- **Design:** Stromal TGF-beta excludes CD8+ T cells and biases T-cell differentiation away from TH1 effectors. Pharmacologic TGF-beta blockade restores T-cell infiltration and unmasks anti-PD-L1 response in tumors that are otherwise immune-cold.
- **n:** n=8-12 mice/arm

- **Effect size:** strong — Anti-PD-L1 alone barely touched liver metastases; galunisertib + anti-PD-L1 produced durable T-cell-mediated regressions and prevented metastatic outgrowth. Foundational preclinical evidence for the TGF-beta + ICI combination strategy relevant to the patient's MSS, SMAD4-mutant, peritoneal-spreading phenotype.
- **Variance:** vs anti-PD-L1 alone; galunisertib alone; vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Liver-metastasis model; peritoneal-spread arm not part of the original Nature paper. Clinical galunisertib + ICI trials in CRC have not yet replicated this depth of response.

Li/Xu (2018) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Fruquintinib
- **Indication / model:** Refractory metastatic colorectal cancer (FRESCO) (3L+); —
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Hyman/Solit (2017) — Cancer Discov

Reference:PMID 28645941 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Capivasertib (AZD5363) phase 2 in AKT1 E17K-mutant solid tumors
- **Indication / model:** Solid tumors with AKT1 E17K mutation (2L+); n=58 AKT1 E17K solid tumors (ER+ breast, gynecologic, other; CRC ~10%).
- **Design:** Phase 2 single-arm basket
- **n:** 58
- **Effect size:** 5.5 months — ORR / PFS by tumor type
- **Variance:** mPFS 5.5 mo (ER+ breast); 6.6 mo (gyn); 4.2 mo (other)
- **Toxicities:** Hyperglycemia G>=3 (24%); Diarrhea G>=3 (17%); Rash G>=3 (15%)
- **Case fit:** weak
- **Notes:** Cross-tumor anchor for capivasertib in AKT-mutant solid tumors — not PIK3CA-specific. Closest published data on capivasertib activity in non-breast solid tumors. The CAPItello-292 / CAPItello-291 CRC + PIK3CA-axis data have not posted as of 2026. Audit-trail row for the AKT-inhibitor class option.

Tanaka/Furukawa (2017) — Mol Cancer Ther

Reference:PMID 28202517 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Tankyrase inhibition in APC-mutant CRC - APC-allele dependence of response
- **Indication / model:** APC-mutant CRC cell-line panel (COLO320, SW480, HT29, HCT116) and xenografts; isogenic APC truncation series
- **Design:** Tankyrase 1/2 PARylates Axin, marking it for degradation. Inhibition stabilizes Axin and restores beta-catenin destruction complex activity - but only when the APC mutation retains enough beta-catenin-binding

20-amino-acid repeats (20-AAR) for the residual destruction complex to assemble.
- **n:** n=8-10 mice/arm; cell panels n=15+
- **Effect size:** moderate — Tankyrase inhibition suppressed colony formation and xenograft growth in COLO-320-class APC truncations with low 20-AAR retention; lines with longer APC truncations and more residual 20-AARs were resistant. mTOR signaling identified as a key resistance node. Explains why APC-mutant CRC has not delivered as clean a tankyrase-inhibitor response as the biology predicted.
- **Variance:** vs APC-WT (HCT116) and vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Patient's APC E1295 sits in the mutation cluster region (codons 1250-1500); the 20-AAR retention will depend on whether E1295 is truncating or missense. Strong preclinical biology, weak clinical translation to date - flag as biology-promising-but-trial-only.

Sartore-Bianchi/Siena (2016) — Lancet Oncol

Reference:PMID 27108243 · **Type:** clinical · **Inclusion:** excluded — Trastuzumab + lapatinib in HER2+ KRAS exon 2 WT mCRC. Patient's KRAS A59T (exon 3) was outside HERACLES eligibility, and HER2 is not on the patient's nominated targetable features. Audit-trail row for HER2 landscape.

- **Intervention:** Trastuzumab + lapatinib (HERACLES, HER2+ KRAS-WT mCRC)
- **Notes:** Excluded: Trastuzumab + lapatinib in HER2+ KRAS exon 2 WT mCRC. Patient's KRAS A59T (exon 3) was outside HERACLES eligibility, and HER2 is not on the patient's nominated targetable features. Audit-trail row for HER2 landscape. — ORR 30%, mPFS 5.2 mo, mOS 11.5 mo. Original HER2-dual-blockade proof-of-concept in CRC.

Roh/Ogino (2016) — Clin Cancer Res

Reference:PMID 26861460 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SMAD4 expression loss as prognostic biomarker in colon cancer
- **Indication / model:** Resected colon cancer (stage I-III predominant) (any); n=1,170 colon cancer specimens, SMAD4 IHC, multivariable Cox adjustment for stage, differentiation, adjuvant therapy, lymphovascular invasion.
- **Design:** Retrospective cohort with biomarker
- **n:** 1170
- **Effect size:** 1.45 HR (7-yr cancer-specific survival, SMAD4-lost) — 5-year PFS and 7-year CSS by SMAD4 expression
- **Variance:** 95% CI 1.06–1.99; 95% CI 1.06-1.99; PFS HR 1.27 (1.01-1.60); p=0.022
- **Toxicities:** Not applicable.
- **Case fit:** partial
- **Notes:** Independently prognostic SMAD4 loss in colon cancer after adjustment for stage and other covariates. Supports the SMAD4 R361H pessimistic-prognostic framing for the patient and the rationale for trial enrichment around TGFb-axis modulation. PMC6755646.

Scherr/Witzigmann (2016) — Mol Cancer

Reference:PMID 26715361 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** BCL-xL as critical survival factor in 20q11-amplified CRC

- **Indication / model:** CRC cell-line panel stratified by 20q11.21 (BCL2L1) amplification; selective BCL-xL inhibitors (A-1331852, navitoclax); xenografts
- **Design:** BCL2L1 amplification at 20q11.21 confers BCL-xL dependence by setting a high threshold for intrinsic-apoptosis activation. BH3-mimetic inhibition of BCL-xL releases pro-apoptotic BIM/BAK/BAX and triggers apoptosis.
- **n:** 20+ CRC lines, xenograft n=8-10/arm
- **Effect size:** strong — BCL2L1-amplified CRC lines were 10-100x more sensitive to BCL-xL-selective inhibitors than non-amplified lines, with xenograft regression and apoptosis induction. Provides the direct preclinical rationale for navitoclax / DT2216 strategies in the patient's BCL2L1-amplified tumor.
- **Variance:** vs BCL2L1-non-amplified CRC lines + vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line and xenograft only; no orthotopic or peritoneal model. Navitoclax thrombocytopenia limits clinical dose and is the main translatability concern.

Ebert/Hoeflich (2016) — Immunity

Reference:PMID 27192565 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MEK inhibition + anti-PD-L1 in CT26 and other MSS-CRC models
- **Indication / model:** CT26 and MC38 syngeneic mouse CRC; cobimetinib + anti-PD-L1 combination; immune profiling
- **Design:** MEK inhibition was thought to upregulate MHC-I, increase intratumoral CD8 T-cell accumulation, and prolong T-cell survival - a 'turn cold to hot' rationale that motivated combining cobimetinib with atezolizumab in MSS CRC.
- **n:** n=10-15 mice/arm
- **Effect size:** moderate — Preclinical combination regressed CT26 and MC38 tumors more durably than either single agent and supported the IMblaze370 phase 3 trial. The clinical trial then failed to improve OS over regorafenib, highlighting the preclinical-to-clinical translation gap in MSS-CRC immunotherapy combinations.
- **Variance:** vs vehicle, MEKi alone, anti-PD-L1 alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Translational failure is the actionable lesson. Listed in audit trail to make the biology-strong / clinical-fail pattern visible. Do not extrapolate uncritically from MC38/CT26 to MSS CRC patients.

Simkens/Punt (2015) — Lancet

Reference:PMID 25862517 · **Type:** clinical · **Inclusion:** excluded — Maintenance backbone after induction CAPOX-bev. Mechanism does not target the patient's features.

- **Intervention:** Capecitabine + bevacizumab maintenance (CAIRO3)
- **Notes:** Excluded: Maintenance backbone after induction CAPOX-bev. Mechanism does not target the patient's features. — Maintenance arm improved PFS1 (11.7 vs 8.5 mo) over observation. Decision-relevant for post-HIPEC maintenance strategy; audit-trail row.

Tabernero/Garcia-Carbonero (2015) — Lancet Oncol

Reference:PMID 25877855 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 2L backbone (ramucirumab

+ FOLFIRI). Anti-VEGFR2 mechanism does not target the patient's features.

- **Intervention:** Ramucirumab + FOLFIRI (RAISE, 2L post-bev)
- **Notes:** Excluded: Standard-of-care 2L backbone (ramucirumab + FOLFIRI). Anti-VEGFR2 mechanism does not target the patient's features. — 2L after bev-containing 1L: mOS 13.3 vs 11.7 mo (HR 0.84, p=0.02). Audit-trail row.

Li/Xu (2015) — Lancet Oncol

Reference:PMID 25981818 · **Type:** clinical · **Inclusion:** excluded — Regorafenib backbone confirmation in Asian patients (CONCUR). Mechanism does not target the patient's features.

- **Intervention:** Regorafenib monotherapy (CONCUR, Asian cohort, 3L+)
- **Notes:** Excluded: Regorafenib backbone confirmation in Asian patients (CONCUR). Mechanism does not target the patient's features. — mOS 8.8 vs 6.3 mo (HR 0.55, p=0.00016). Replicates CORRECT in Asian population.

Mayer/Ohtsu (2015) — N Engl J Med

Reference:PMID 25970050 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 3L backbone (TAS-102 monotherapy). Mechanism does not target the patient's features.

- **Intervention:** Trifluridine/tipiracil (TAS-102) monotherapy (RECOURSE, 3L+)
- **Notes:** Excluded: Standard-of-care 3L backbone (TAS-102 monotherapy). Mechanism does not target the patient's features. — Pivotal: mOS 7.1 vs 5.3 mo (HR 0.68, p<0.001). Audit-trail row.

Sorich/Karapetis (2015) — Ann Oncol

Reference:PMID 25115304 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Anti-EGFR mAbs in extended-RAS-mutant mCRC (meta-analysis)
- **Indication / model:** Metastatic colorectal cancer stratified by extended-RAS status (KRAS exons 2/3/4 and NRAS exons 2/3/4) (any); Pooled across CRYSTAL, PRIME, OPUS, FIRE-3, PEAK, 20020408, 20050181, NORDIC-VII, MRC COIN — 5,948 patients; ~20% of KRAS exon-2 WT tumors carried a 'new RAS' alteration.
- **Design:** Systematic review and meta-analysis of randomised controlled trials
- **n:** 5948
- **Effect size:** 1.08 HR (any new RAS mutation, OS) — OS / PFS hazard ratios for anti-EGFR therapy by extended-RAS status
- **Variance:** 95% CI 0.91–1.28; 95% CI 0.91-1.28 (OS in new-RAS-mut subset); PFS HR 1.03 (0.78-1.36)
- **Toxicities:** Class-typical anti-EGFR rash and hypomagnesemia plus chemo toxicity; safety was not the focus of this meta-analysis.
- **Case fit:** partial
- **Notes:** Foundational evidence that non-canonical RAS mutations (which include codon 59) abrogate the anti-EGFR survival benefit at the population level. The Lochhead A59T case report sits against this signal, not with it.

Schirripa/Loupakis (2015) — Int J Cancer

Reference:PMID 24806288 · **Type:** clinical · **Inclusion:** included

- **Intervention:** NRAS / atypical RAS mutations as anti-EGFR resistance markers (retrospective cohort)
- **Indication / model:** Metastatic colorectal cancer, chemo-refractory (2L+); n=786 mCRC patients; NRAS mutated in 47 (6%). NRAS-mut vs KRAS-mut groups compared for clinicopathology and anti-EGFR response.
- **Design:** Retrospective single-institution cohort
- **n:** 786
- **Effect size:** 0 % — Median OS and anti-EGFR response rate by RAS status
- **Variance:** 0/9 evaluable NRAS-mut patients responded to anti-EGFR
- **Toxicities:** Retrospective; safety not assessed as a primary endpoint.
- **Case fit:** partial
- **Notes:** Adjacent biology — extends the 'atypical-RAS-blocks-anti-EGFR' signal to NRAS. The patient's KRAS A59T sits in the same conceptual family. Reinforces the Sorich meta-analytic call.

Vlahovic/Tolcher (2015) — Cancer Chemother Pharmacol

Reference:PMID 26429709 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Navitoclax (ABT-263) + irinotecan phase 1 in solid tumors
- **Indication / model:** Advanced solid tumors (mixed; CRC subset present) (2L+); n=39 advanced solid tumors. UGT1A1*28 7/7 homozygotes excluded by amendment due to G3+ diarrhea / febrile neutropenia signal.
- **Design:** Phase 1 dose-escalation, single-arm
- **n:** 39
- **Effect size:** 1 PR per arm — — Safety, RP2D, anti-tumor activity
- **Variance:** RP2D defined; PR in 1 patient each schedule
- **Toxicities:** Diarrhea G>=3 (35%); Thrombocytopenia G>=3 (24%); Neutropenia G>=3 (18%); Diarrhea (84%)
- **Case fit:** weak
- **Notes:** Most directly applicable precedent for the patient's BCL2L1 (BCL-xL) + TOP1 co-amplification biology — navitoclax + irinotecan was tolerable at the published RP2D, with on-target thrombocytopenia driving the field's move to DT2216 PROTACs. Modest activity (5% ORR overall) in unselected solid tumors; selection for BCL2L1-amp tumors was not tested.

Le/Diaz (2015) — N Engl J Med

Reference:PMID 26028255 · **Type:** clinical · **Inclusion:** excluded — Pembrolizumab in MMR-deficient solid tumors (Le 2015). Patient is MSS / TMB-low / PD-L1-neg by intake — the opposite of the Le 2015 responder phenotype. Listed for completeness because this paper anchors the MSI-H exception that the patient does NOT have.

- **Intervention:** Pembrolizumab in MMR-deficient mCRC (Le et al.)
- **Notes:** Excluded: Pembrolizumab in MMR-deficient solid tumors (Le 2015). Patient is MSS / TMB-low / PD-L1-neg by intake — the opposite of the Le 2015 responder phenotype. Listed for completeness because this paper anchors the MSI-H exception that the patient does NOT have. — Landmark dMMR / pMMR contrast — ORR 40% (dMMR CRC) vs 0% (pMMR CRC). Defines why MSS-CRC remains ICI-cold.

Guinney/Tejpar (2015) — Nat Med

Reference:PMID 26457759 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Consensus Molecular Subtype CMS4 mesenchymal CRC - SMAD4/TGF-beta biology

- **Indication / model:** Integration of 6 CRC transcriptomic classifications across ~4,000 samples; PDX and cell-line CMS calls in follow-on work
- **Design:** CMS4 (~23% of CRC) is the mesenchymal subtype - stromal infiltration, TGF-beta activation, EMT signature, and worst prognosis. SMAD4 loss and peritoneal involvement are enriched.
- **n:** n=4,151 CRC transcriptomes
- **Effect size:** moderate — CMS4 carries the worst overall and relapse-free survival across stages, and is enriched for stromal TGF-beta signaling, SMAD4 alterations, and EMT features. Frames the patient's SMAD4-R361H, peritoneal-spread, CDX2-unknown phenotype as likely CMS4 with the prognostic and therapeutic implications that follow.
- **Variance:** vs CMS1/2/3 contrasts; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Taxonomic / prognostic - not a treatment-effect study. CMS4 prognosis worsens further when combined with SMAD4 loss and peritoneal disease.

Tabernero/Garcia-Carbonero (2015) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Ramucirumab + FOLFIRI
- **Indication / model:** Second-line metastatic colorectal cancer (RAISE) (2L); —
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Heinemann/Stintzing (2014) — Lancet Oncol

Reference:PMID 25088940 · **Type:** clinical · **Inclusion:** excluded — FIRE-3 cetuximab vs bevacizumab in RAS-WT 1L mCRC — the patient has KRAS A59T (atypical RAS-mutated), so the RAS-WT subset that drives the survival signal does not include them. Both arms are SoC backbones not targeting the patient's specific features. Atypical-KRAS / anti-EGFR question is covered separately in Bucket H rows.

- **Intervention:** FOLFIRI + cetuximab vs FOLFIRI + bevacizumab (FIRE-3)
- **Notes:** Excluded: FIRE-3 cetuximab vs bevacizumab in RAS-WT 1L mCRC — the patient has KRAS A59T (atypical RAS-mutated), so the RAS-WT subset that drives the survival signal does not include them. Both arms are SoC backbones not targeting the patient's specific features. Atypical-KRAS / anti-EGFR question is covered separately in Bucket H rows. — RAS-WT 1L mCRC: cetuximab arm mOS 28.7 vs 25.0 mo, HR 0.77. Anchor for the cetuximab-vs-bev comparison; KRAS A59T patient is not RAS-WT by current consensus.

Loupakis/Falcone (2014) — N Engl J Med

Reference:PMID 25337750 · **Type:** clinical · **Inclusion:** excluded — Intensive triplet chemo + bev (FOLFOXIRI-bev) is a backbone-intensification option — mechanism does not target the patient's nominated targetable features. Worth surfacing to the treating team as a 1L alternative; Libby does not rank it.

- **Intervention:** FOLFOXIRI + bevacizumab vs FOLFIRI + bevacizumab (TRIBE)

- **Notes:** Excluded: Intensive triplet chemo + bev (FOLFOXIRI-bev) is a backbone-intensification option — mechanism does not target the patient's nominated targetable features. Worth surfacing to the treating team as a 1L alternative; Libby does not rank it. — Intensive triplet improved mPFS (12.1 vs 9.7 mo, HR 0.75) and mOS in updated analysis (29.8 vs 25.8 mo, HR 0.80). Relevant if treating team considers re-induction in young ECOG-1 patient.

Solass/Reymond (2014) — Ann Surg Oncol

Reference:PMID 24682360 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PIPAC - pressurized intraperitoneal aerosol chemotherapy tissue distribution
- **Indication / model:** Ex-vivo human peritoneum; postmortem porcine and sheep models; in vivo porcine PIPAC with doxorubicin/cisplatin
- **Design:** Aerosolization of cytotoxic agents into the pneumoperitoneum produces homogeneous distribution and 3-5x deeper tumor-tissue penetration than static intraperitoneal lavage, exploiting hydrostatic pressure to drive drug into tumor nodules.
- **n:** ex vivo and in vivo porcine n=6-10
- **Effect size:** moderate — PIPAC achieved deeper and more uniform tumor distribution than conventional liquid IP delivery; subsequent organoid work (2025) confirmed enhanced cytotoxicity of PIPAC-delivered cisplatin/oxaliplatin against CRC patient-derived organoids. Underpins the PIPAC arm on the patient's trial list as a complementary strategy if peritoneal disease persists post-HIPEC.
- **Variance:** vs static liquid intraperitoneal chemo (lavage); translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Porcine/sheep + ex-vivo data; in-human PIPAC efficacy data are mainly observational. Repeated PIPAC may have a tolerable safety profile post-CRS-HIPEC.

Sun/Su (2014) — Cancer Biol Med

Reference:PMID 26464702 · **Type:** preclinical · **Inclusion:** excluded — Mechanism (VEGFR-1/2/3 inhibition) is anti-angiogenic and does not target the patient's nominated targetable features. Standard / refractory-line agent run by the treating team; out of Libby feature-targeting scope.

- **Intervention:** Fruquintinib selective VEGFR-1/2/3 inhibitor (Sun et al.)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Mechanism (VEGFR-1/2/3 inhibition) is anti-angiogenic and does not target the patient's nominated targetable features. Standard / refractory-line agent run by the treating team; out of Libby feature-targeting scope.

Wilhelm/Zopf (2014) — Int J Cancer

Reference:PMID 21170960 · **Type:** preclinical · **Inclusion:** excluded — Multikinase angiogenesis inhibitor; mechanism does not target nominated features. Standard refractory-line therapy reviewed for completeness, then excluded from feature-targeting synthesis.

- **Intervention:** Regorafenib multikinase preclinical CRC activity (Wilhelm / Schmieder)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Multikinase angiogenesis inhibitor; mechanism does not target nominated features. Standard

refractory-line therapy reviewed for completeness, then excluded from feature-targeting synthesis.

Tanaka/Matsuoka (2014) — Oncol Rep

Reference:PMID 25117552 · **Type:** preclinical · **Inclusion:** excluded — Trifluridine/tipiracil is a fluoropyrimidine-class antimetabolite; mechanism does not engage any nominated targetable feature. Standard later-line backbone.

- **Intervention:** TAS-102 (trifluridine/tipiracil) preclinical mechanism
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Trifluridine/tipiracil is a fluoropyrimidine-class antimetabolite; mechanism does not engage any nominated targetable feature. Standard later-line backbone.

AbbVie/AbbVie (2014) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Navitoclax (ABT-263) monotherapy or combined with irinotecan or erlotinib
- **Indication / model:** Advanced solid tumors (2L+); —
- **Design:** 1
- **Effect size:** — — Safety, RP2D
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Grothey/Van Cutsem (2013) — Lancet

Reference:PMID 23177514 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 3L backbone (regorafenib monotherapy). Multikinase VEGFR/TIE2 mechanism does not target the patient's features.

- **Intervention:** Regorafenib monotherapy (CORRECT, 3L+)
- **Notes:** Excluded: Standard-of-care 3L backbone (regorafenib monotherapy). Multikinase VEGFR/TIE2 mechanism does not target the patient's features. — Pivotal: mOS 6.4 vs 5.0 mo (HR 0.77, p=0.0052). High G3+ AE rate (hand-foot 17%, fatigue 10%, HTN 7%). Audit-trail row.

Liu/Cong (2013) — Proc Natl Acad Sci USA

Reference:PMID 23258887 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** WNT974/LGK974 porcupine inhibitor - RNF43-mutant dependence
- **Indication / model:** RNF43-mutant pancreatic and CRC organoids; xenografts; PORCN inhibitor-sensitivity profiling
- **Design:** PORCN (porcupine) acylates Wnt ligands to enable secretion. Inhibition starves ligand-dependent tumors of Wnt signaling - effective when the tumor depends on autocrine/paracrine Wnt ligand (RNF43-loss-of-function, RSPO fusion), but bypassed in APC-mutant tumors where the downstream destruction complex is already inactivated.
- **n:** panel screens + n=8/arm xenograft
- **Effect size:** strong — LGK974 caused tumor regression in RNF43-mutant and RSPO-fusion-positive models,

but did not affect APC-mutant CRC because APC truncation acts downstream of secreted Wnt. Frames the patient's APC E1295 as a likely resistance feature for porcupine inhibitor strategies.
- **Variance:** vs RNF43 WT + APC-mutant CRC + vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Confirms APC-mutant CRC sits outside the PORCN-inhibitor sweet spot - useful to caveat the patient's APC mutation as a negative predictor for WNT974, not a positive one.

Itatani/Taketo (2013) — Gastroenterology

Reference:PMID 23891973 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** SMAD4 loss drives CCL15-CCR1 myeloid recruitment and liver metastasis
- **Indication / model:** Smad4-conditional Apc-mutant mouse CRC; orthotopic and spleen-injection liver metastasis; human CRC tissue correlation
- **Design:** SMAD4 loss derepresses CCL15 expression in CRC cells, recruiting CCR1+ myeloid cells to liver-metastasis niches; CCR1 antagonism blunts metastasis in vivo. Parallel CCL20-CCR6 and other chemokine axes also drive peritoneal-omental homing in subsequent literature.
- **n:** n=6-10 mice/arm; ~100 human tissues
- **Effect size:** moderate — SMAD4-null tumors had higher CCL15, more CCR1+ myeloid infiltrate, and more liver metastases than SMAD4-WT controls; pharmacologic CCR1 blockade reduced metastatic burden. Links SMAD4 loss directly to a targetable myeloid axis relevant to peritoneal and hepatic spread.
- **Variance:** vs Smad4-WT Apc-mutant littermates; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Liver-focused; peritoneal homing data are extrapolated from related CCL/CCR axes. CCR1 antagonists are not clinically available in oncology.

Van Cutsem/Allegra (2012) — J Clin Oncol

Reference:PMID 22949147 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 2L backbone (aflibercept + FOLFIRI). Anti-angiogenic mechanism does not target the patient's nominated features.

- **Intervention:** Aflibercept + FOLFIRI (VELOUR, 2L)
- **Notes:** Excluded: Standard-of-care 2L backbone (aflibercept + FOLFIRI). Anti-angiogenic mechanism does not target the patient's nominated features. — 2L after oxaliplatin failure: mOS 13.5 vs 12.1 mo (HR 0.82, p=0.003). Anchors aflibercept; only relevant if sequence switches to FOLFOX 1L.

Liao/Ogino (2012) — N Engl J Med

Reference:PMID 23094721 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Regular aspirin use in PIK3CA-mutant CRC (Nurses' Health + Health Professionals cohorts)
- **Indication / model:** Resected colorectal cancer stratified by tumor PIK3CA mutation (adj); 964 stage I-IV CRC patients from NHS + HPFS; tumor PIK3CA mutation status assessed retrospectively; 17% PIK3CA-mut.
- **Design:** Prospective cohort study (post-hoc biomarker analysis)
- **n:** 964
- **Effect size:** 0.18 HR (CRC-specific mortality, PIK3CA-mut subset) — CRC-specific and overall mortality by

aspirin use × PIK3CA status
- **Variance:** 95% CI 0.06–0.61; 95% CI 0.06-0.61; p=0.001
- **Toxicities:** Aspirin GI-bleed / ulcer risk class-typical; not formally captured in this cohort analysis.
- **Case fit:** partial
- **Notes:** Foundational observational signal that aspirin halves CRC-specific mortality in PIK3CA-mut patients. The patient's metastatic setting differs from the resected adjuvant cohort here, but the biomarker direction matches.

Lehmann/Mohell (2012) — J Clin Oncol

Reference:PMID 22965953 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Eprenetapopt (APR-246) phase 1 in TP53-mutant solid tumors and hematologic malignancies
- **Indication / model:** Solid tumors and hematologic malignancies with TP53 mutation (2L+); n=22, mixed solid + heme; first-in-human PK and tolerability study.
- **Design:** Phase 1 single-arm dose-escalation
- **n:** 22
- **Effect size:** RP2D 67 mg/kg/day — — Safety, RP2D, biological activity (p53 reactivation by transcriptional readout)
- **Variance:** no DLTs to top dose; PD activity by gene-expression panel
- **Toxicities:** Fatigue (50%); Nausea (45%); Dizziness/confusion G1-2 (32%)
- **Case fit:** weak
- **Notes:** Lehmann/Mohell phase 1 — first APR-246 readout established the agent's tolerability and biological activity but did not establish solid-tumor efficacy. Aprea pivoted to MDS/AML, then failed phase 3 in TP53-mut MDS (2020). The TP53-mut solid-tumor combination studies (NCT04383938 APR-246 + pembrolizumab) ran without clear positive signal.

Davies/Pass (2012) — Mol Cancer Ther

Reference:PMID 22294718 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Capivasertib (AZD5363) pan-AKT kinase inhibitor
- **Indication / model:** Broad cell-line panel including PIK3CA-mutant CRC and breast lines; xenograft series
- **Design:** ATP-competitive pan-AKT (AKT1/2/3) inhibitor that blunts PI3K/AKT/mTOR signaling downstream of activated p110-alpha or PTEN loss.
- **n:** n=8-10/arm xenograft groups
- **Effect size:** moderate — Single-agent capivasertib produced regressions in PIK3CA-mutant breast and prostate xenografts at well-tolerated doses, with weaker monotherapy activity in PIK3CA-mutant CRC unless combined with chemo or anti-EGFR. Supports the AKT axis as a node, with the caveat that KRAS co-activation dampens monotherapy response.
- **Variance:** vs vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Decade-old preclinical panel; CAPItello-291 in breast cancer carries the modern clinical read-through. CRC PDX activity has been modest as monotherapy.

Liao/Ogino (2012) — N Engl J Med

Reference:PMID 23094721 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Aspirin chemoprevention in PIK3CA-mutant CRC - mechanism
- **Indication / model:** Retrospective clinical cohorts (NHS, HPFS) totaling ~964 CRC patients; in vitro PIK3CA-mutant CRC cells with aspirin exposure
- **Design:** PIK3CA mutation activates the COX-2 / PGE2 / PI3K axis; aspirin inhibits COX-2-driven PGE2 and indirectly dampens PI3K signaling, with greater anti-proliferative activity in PIK3CA-mutant CRC cells than WT.
- **n:** 964 patients + in vitro panel
- **Effect size:** moderate — Post-diagnosis aspirin associated with markedly improved cancer-specific and overall survival in PIK3CA-mutant CRC (HR ~0.18) but no benefit in WT. Preclinical follow-on confirmed greater anti-proliferative effect of aspirin in PIK3CA-mutant CRC lines via PI3K/AKT/mTOR modulation.
- **Variance:** vs PIK3CA-WT CRC and non-aspirin users; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Original Liao paper is observational, but the PIK3CA-stratified preclinical rationale and the recent ALASCCA RCT (2025) provide stronger causal support. Risk/benefit for the patient depends on bleeding risk, planned surgery timing, and bevacizumab co-administration.

Van Cutsem/Van Cutsem (2012) — ClinicalTrials.gov registration

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Aflibercept + FOLFIRI
- **Indication / model:** Oxaliplatin-pretreated mCRC (VELOUR) (2L); —
- **Design:** 3
- **Effect size:** — — OS
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Gandhi/Tolcher (2011) — J Clin Oncol

Reference:PMID 21282543 · **Type:** clinical · **Inclusion:** included

- **Intervention:** Navitoclax (ABT-263) monotherapy phase 1 in SCLC and other solid tumors
- **Indication / model:** Small-cell lung cancer and other advanced solid tumors (2L+); n=47 (SCLC predominant); other solid tumors included.
- **Design:** Phase 1 dose-escalation, single-arm
- **n:** 47
- **Effect size:** 2.5 % — Safety, RP2D, ORR
- **Variance:** 1 PR in SCLC; SD in additional patients
- **Toxicities:** Thrombocytopenia G>=3 (28%); Diarrhea (40%); Fatigue (45%)
- **Case fit:** weak
- **Notes:** Foundational navitoclax monotherapy paper — establishes the class-defining platelet toxicity that motivated DT2216 PROTAC design. CRC subset not separately analyzed.

Reaper/Pollard (2011) — Nat Chem Biol

Reference:PMID 21814628 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition (AZD6738/ceralasertib) synthetic lethality in TP53-defective tumors
- **Indication / model:** TP53-null and TP53-mutant cell line panels including CRC; xenograft series
- **Design:** ATR is the principal sensor of replication-stress and ssDNA. Loss of TP53 forces cells onto ATR-CHK1-driven G2/M control of replication-stress; ATR inhibition then triggers replication catastrophe selectively in TP53-mutant cells. Effect amplified by chemotherapy that increases replication stress (irinotecan).
- **n:** panel screens + n=8-10/arm xenograft
- **Effect size:** moderate — AZD6738 selectively sensitized TP53-mutant CRC and CLL cells to chemo (cisplatin, irinotecan, carboplatin) with strong synthetic lethality. Provides the mechanistic basis for combining ceralasertib/camonsertib with the patient's existing irinotecan backbone or HIPEC platinum.
- **Variance:** vs TP53 WT isogenic + vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Original work was CHK1/ATR axis, multiple tumor types. CRC-specific in vivo regression is largely cell-line xenograft. Hematologic toxicity has limited dose intensity in clinical TP53-mutant combos.

Van der Speeten/Sugarbaker (2010) — Cancer Chemother Pharmacol

Reference:PMID 20689948 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Mitomycin C HIPEC pharmacokinetics in colorectal peritoneal metastasis
- **Indication / model:** Clinical pharmacokinetic study in 145 patients undergoing CRS + MMC HIPEC; supporting in vitro tumor-tissue penetration models
- **Design:** Peritoneal-plasma barrier creates a 20-30x concentration advantage for hydrophilic agents like MMC in the peritoneal cavity. Hyperthermia enhances tumor tissue penetration (~3-5 mm) and DNA-crosslink formation. Pharmacokinetic profile is markedly altered by patient surface area, perfusate volume, and intraoperative variables.
- **n:** n=145 patients
- **Effect size:** strong — Peritoneal-to-plasma MMC area-under-curve ratio of ~20-25, with intraperitoneal MMC clearance dominated by tissue uptake rather than systemic absorption. Justifies HIPEC as a concentration-x-time advantage strategy and underpins MMC dosing for the patient's planned procedure.
- **Variance:** vs intravenous MMC reference pharmacokinetics; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Clinical pharmacokinetic study (not strictly preclinical); included because mechanism = pharmacology and the patient's HIPEC plan depends on this concentration-time logic. Neutropenia ~25-40% on weight-based dosing is the main toxicity.

Rangel-Moreno/Randall (2010) — J Immunol

Reference:PMID 20498369 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Omental milky spots as a metastatic and immune niche in the peritoneal cavity
- **Indication / model:** Mouse peritoneal cavity / omentum immunohistology; tumor-cell seeding; B1 lymphocyte and DC characterization
- **Design:** Omental milky spots are organized immune aggregates rich in B1 lymphocytes, peritoneal macrophages, and DCs. They serve as the primary lymphoid niche for the peritoneal cavity but are skewed

away from canonical antitumor adaptive responses, biasing the peritoneal microenvironment toward myeloid-suppressed states.
- **n:** n=5-10 mice per experiment
- **Effect size:** moderate — Omental milky spots preferentially trap disseminating tumor cells and modulate local T-cell priming; the peritoneal cavity is dominated by B1 cells rather than conventional B2 follicular cells. Helps explain why systemic ICI underperforms in peritoneal-dominant disease relative to liver-only disease - the immune niche is biologically different.
- **Variance:** vs splenectomized + lymph-node-deficient comparators; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mouse immune anatomy generalizes imperfectly to human omentum, but the principle - peritoneal cavity is an immunologically distinct compartment - is supported across species. Argues for loco-regional immunotherapy delivery strategies on top of systemic ICI.

Hirai/Iwasawa (2009) — Mol Cancer Ther

Reference:PMID 19887545 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** WEE1 inhibition (adavosertib) in RAS + TP53 co-mutant CRC - preclinical rationale
- **Indication / model:** TP53-mutant CRC and ovarian cell lines and xenografts; isogenic TP53 KO panels
- **Design:** WEE1 inhibition forces premature mitotic entry in cells already dependent on G2/M arrest due to loss of G1/S TP53 checkpoint. Synthetic-lethal in TP53-mutant cells, amplified in RAS co-activated context where replication stress is elevated.
- **n:** panels of 20+ lines; xenograft n=8-10/arm
- **Effect size:** moderate — Adavosertib produced stronger growth arrest and apoptosis in TP53-mutant lines than isogenic WT controls, with combination synergy with gemcitabine, irinotecan, or carboplatin. Forms the rationale for the FOCUS4-C clinical trial in RAS+TP53-mutant mCRC (PFS 3.61 vs 1.87 mo).
- **Variance:** vs TP53 WT isogenic + vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Earlier mechanism papers came from ovarian/HNSCC; CRC-specific PDX data is thinner. Toxicity in early CRC combination trials has limited dose.

Saltz/Cassidy (2008) — J Clin Oncol

Reference:PMID 18421054 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 1L backbone (FOLFOX/XELOX + bevacizumab). Mechanism does not target the patient's nominated features.

- **Intervention:** Bevacizumab + FOLFOX-4 or XELOX (NO16966)
- **Notes:** Excluded: Standard-of-care 1L backbone (FOLFOX/XELOX + bevacizumab). Mechanism does not target the patient's nominated features. — 1L oxaliplatin + bev RCT (mPFS 9.4 vs 8.0 mo, HR 0.83). Backbone evidence retained in the audit trail.

Fuchs/Mitchell (2007) — J Clin Oncol

Reference:PMID 17947725 · **Type:** clinical · **Inclusion:** excluded — Backbone chemotherapy comparison (FOLFIRI vs mIFL vs CapeIRI) — mechanism does not target the patient's nominated features.

- **Intervention:** FOLFIRI vs mIFL vs CapeIRI (BICC-C)
- **Notes:** Excluded: Backbone chemotherapy comparison (FOLFIRI vs mIFL vs CapeIRI) — mechanism does not target the patient's nominated features. — Established FOLFIRI as the favored 1L irinotecan-fluoropyrimidine schedule (mOS 23.1 mo vs 17.6/18.9 mo). Audit-trail row.

Hurwitz/Kabbinavar (2004) — N Engl J Med

Reference:PMID 15175435 · **Type:** clinical · **Inclusion:** excluded — Standard-of-care 1L backbone (IFL/FOLFIRI + bevacizumab) whose mechanism (VEGF-A neutralization) does not target the patient's nominated targetable features (KRAS A59T, PIK3CA M1043I, APC E1295, TP53 R273, SMAD4 R361H, BCL2L1/TOP1 amp). Libby is feature-targeting scope only; SoC backbones run through the treating team independent of this dossier.

- **Intervention:** Bevacizumab + IFL (AVF2107g)
- **Notes:** Excluded: Standard-of-care 1L backbone (IFL/FOLFIRI + bevacizumab) whose mechanism (VEGF-A neutralization) does not target the patient's nominated targetable features (KRAS A59T, PIK3CA M1043I, APC E1295, TP53 R273, SMAD4 R361H, BCL2L1/TOP1 amp). Libby is feature-targeting scope only; SoC backbones run through the treating team independent of this dossier. — Anchor RCT for the bevacizumab-plus-chemo backbone in 1L mCRC (mOS 20.3 vs 15.6 mo, HR 0.66, p<0.001). Listed for audit trail; not part of Libby's feature-targeting synthesis.